

PREDICTIVE LEARNING FOR CHILI VARIETIES FROM MULTI-SOURCE IMAGE AND SENSOR DATA

An INDUSTRIAL ORIENTED MINI PROJECT REPORT

Submitted by

K. KUSUMA (23UJ1A6673)
M. SRAVANTHI (23UJ1A6678)
P. YES KUMAR (23UJ1A6687)
R. GANDHI NAIK (23UJ1A6693)

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING (AI&ML)

Under the Guidance of

Mrs. P. ASHWINI M.Tech.
Assistant Professor



**MALLA REDDY ENGINEERING COLLEGE AND
MANAGEMENT SCIENCES**

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Kistapur (V),Medchal(Dt)-501401

Department of Computer Science and Engineering (AI&ML)
Malla Reddy Engineering College and Management Sciences

BONAFIDE CERTIFICATE

This is to certify that the project report titled, “ Predictive Learning For Chili Varieties From Multi-Source Image And Sensor Data ” is a bonafide record of the project work done by

K. KUSUMA (23UJ1A6673)
M. SRAVANTHI (23UJ1A6678)
P. YES KUMAR (23UJ1A6687)
R. GANDHI NAIK (23UJ1A6693)

The award of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence &Machine Learning) during the academic year 2025-2026.

Head of the Department Signature: Name: Mr.E.BABU Seal: Date:	Guide Signature: Name: Mrs.P.ASHWINI Designation: Assistant Professor Date:
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Submitted for the University Examination held on

Internal Examiner Signature: Name: Designation: Assistant Professor Date:	External Examiner Signature: Name: College / Designation: Date:
--	--

DECLARATION

We, undersigned, hereby declare that the project report “**Predictive Learning for Chili Varieties using Multi-Source Image and Sensor Data**”, submitted for partial fulfillment of the requirements for the award of degree of **Bachelor of Technology of the Jawaharlal Technological University, Hyderabad** is a bonafide work done by me under the guidance of Mrs.P.Ashwini,M.Tech , Department of CSE(AI&ML).

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Date of Submission:

(Signature)

Name of the students

K. KUSUMA	(23UJ1A6673)
M. SRAVANTHI	(23UJ1A6678)
P. YES KUMAR	(23UJ1A6687)
R. GANDHI NAIK	(23UJ1A6693)

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Name of the students.

K. KUSUMA	(23UJ1A6673)
M. SRAVANTHI	(23UJ1A6678)
P. YES KUMAR	(23UJ1A6687)
R. GANDHI NAIK	(23UJ1A6693)

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CO3	Design engineering solutions to complex problems utilizing a systems approach.
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CO5	Demonstrate the knowledge and skills acquired during the course work

PREDICTIVE LEARNING FOR CHILI VARIETIES FROM MULTI-SOURCE IMAGE AND SENSOR DATA

ABSTRACT

Chili cultivation is a vital component of agricultural production and Agri-based industries, where accurate identification of chili varieties and quality conditions directly influences market value, processing efficiency, and export standards. With the growing scale of cultivation and the availability of digital imaging devices, large volumes of chili images are generated from farms, markets, and storage units. However, effective analysis of such multi-source image data remains a challenge. Accurate and automated classification is therefore essential to support quality assessment and decision-making in modern agricultural systems. Traditionally, chili classification and quality evaluation are performed through manual visual inspection by farmers or quality experts. This traditional system relies heavily on human experience and subjective judgment based on color, shape, texture, and visible defects. Such methods are time-consuming, labor-intensive, and inconsistent, especially when dealing with large datasets or visually similar chili varieties. The major limitations of the traditional approach include lack of standardization, higher chances of human error, difficulty in handling large-scale data, and reduced reliability under varying environmental conditions. To overcome these limitations, there is a strong need for an automated, objective, and scalable image-based classification system. This project proposes a machine learning-based Chili Multi-Source Image Classification framework that utilizes image preprocessing, feature vector generation, and supervised learning algorithms to accurately categorize chili images. The proposed system evaluates multiple classifiers, including Light Gradient Boosting Machine, Linear Discriminant Analysis, AdaBoost, and Support Vector Machine, to identify the most effective model for classification. Experimental results demonstrate significant performance improvement over traditional methods, with the LGBM classifier achieving an accuracy of 98.93% along with high precision, recall, and F1-score values. The significance of the proposed system lies in its ability to reduce human dependency, improve classification consistency, and support real-time prediction through model reuse. The system offers a reliable and efficient solution for automated chili quality assessment, contributing to reduced post-harvest losses, enhanced supply-chain efficiency, and scalable agricultural decision-support systems.

Keywords: Feature Vector Generation, Automated Quality Assessment, Agricultural Decision-Support Systems, Post-Harvest Loss Reduction, Supply-Chain Efficiency

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LIST OF ABBREVIATIONS

SVM	Support vector machine
LDA	Linear discriminant Analysis
LGBM	Light gradient boosting machine
CV	Computer vision

CHAPTER 1

INTRODUCTION

1.1 Introduction to the Project

Chili s are important agricultural products and economic crops. China is a major producer of chili s, with the planting area accounting for 8–10% of the national vegetable planting area annually, with an output value of approximately 250 billion yuan, providing significant economic and social benefits. Currently, all links in the chili industry, except for harvesting, have been mechanized. Faced with social issues such as an aging population and labor shortages, the development of intelligent harvesting equipment is particularly important. With the advancement of information technology, traditional agriculture is moving towards intelligent development, providing a direction for the upgrading of the chili industry. Target detection of chili fruits can provide necessary technical support and intelligent solutions for the chili harvesting process.



Fig. 1.1: Chilli economic crop varieties.

Early detection of s mainly used traditional machine vision. extracted color and shape features of dried s using the line sectioning method and minimum bounding rectangle algorithm, proposed segmentation of multiple dried images based on the watershed algorithm, and used the naive Bayes algorithm for grading multiple dried s, achieving a classification accuracy of over 95%. used color and texture feature methods to identify the total content of carotene in s, with a mean square error (MSE) accuracy of 80%. adopted digital image processing technology to accurately classify the quality of dried s based on physical features such as length and color of the s, achieving a classification accuracy of 94.4% used

RGB, HSI, and color model features to identify four different varieties of s with and without stems. The recognition rate of s with stems based on RGB color features was 70%, while for s without stems was 85%. The recognition rate based on HSI color features for s with stems was 80%, and for s without stems was 90%. The recognition rate based on color features for s with stems was 85%, and for s without stems was 95% . From the above research, it can be seen that traditional digital image processing techniques often require manual feature extraction in object recognition, which is time-consuming and labor-intensive. Additionally, these techniques are susceptible to background, lighting, and human factors, leading to lower recognition accuracy.

1.2 Background of the Study

In recent years, object detection has mainly focused on grading and sorting processes. The identification of fruits under natural conditions is challenging due to complex environments, diverse lighting conditions, and significant occlusion. With the development of the Internet of Things, big data, AI, and other technologies, non-contact object detection based on machine vision and deep learning has laid the foundation for intelligent harvesting of agricultural crops . There are two main methods of determining the state of crops using computer vision technology. The most frequently used method is determining the input image after converting the result for the last layer through several convolution layers to a value between 0 and 1 through an activation function . Furthermore, based on the autoencoder structure, image reconstruction is performed with an image that follows a normal data distribution using only normal data, and the state of the crop can be determined using errors in the input/output results. However, the image reconstruction method is relatively low in accuracy because it is difficult to reconstruct accurately. Attempts have been made to apply autoencoders to to overcome these shortcomings, yet significant performance improvements have not been achieved . However, such efforts have laid the groundwork for further research to enhance performance through architectural modifications or by proposing new processes. Due to the limitation in accuracy, although performance improvement has been achieved more than before, recent researchers have made significant performance improvements using latent vectors between the encoder and decoder.

1.3 Problem Statement

Chili quality assessment using visual inspection faces several challenges due to high intra-class variation and inter-class similarity among different chili types and conditions. Variations in size, shape, color intensity, surface texture, and drying levels make accurate classification difficult using manual observation. The presence of unhealthy or damaged chilies mixed with healthy samples further complicates the inspection process, especially when large quantities must be evaluated quickly.

Multi-source chili image datasets introduce additional complexity because images are captured under diverse illumination conditions, camera resolutions, and backgrounds. Noise, shadows, occlusions, and inconsistent framing degrade visual quality and reduce the effectiveness of traditional image processing approaches. Manual grading methods are subjective and depend heavily on human expertise, leading to inconsistencies across inspectors and locations.

The absence of automated, standardized visual classification systems results in inefficiencies in sorting, grading, and quality control. This leads to increased labor costs, delayed processing, reduced export quality compliance, and higher rejection rates. Addressing these challenges requires a systematic computational approach capable of handling large-scale, diverse chili image datasets with high accuracy and reliability.

1.4 Objectives of the Project

Modern agri-based companies, food exporters, spice processing industries, and cold storage facilities increasingly rely on data analytics to maintain product quality and meet international standards. Companies involved in chili export face strict grading requirements related to color uniformity, surface defects, contamination, and spoilage levels. In real-time operations, manual inspection fails to keep pace with high throughput sorting lines, leading to delayed processing, inconsistent grading, and economic losses. This creates a strong motivation to adopt automated data-driven analysis methods that can operate reliably at scale. Another major motivation arises from the diversity of chili data sources. Images are captured using mobile devices, industrial cameras, drones, and laboratory imaging systems under varying lighting and background conditions. Such multi-source data introduces high variability, making traditional rule-based image processing techniques unreliable.

Data analysis and machine learning techniques provide the capability to learn discriminative patterns directly from heterogeneous image data, enabling robust classification across different acquisition environments without extensive manual tuning. Agri-tech startups and large-scale food companies are integrating analytics pipelines to monitor crop quality from farm to market. Effective data analysis helps reduce post-harvest losses, improves supply-chain transparency, supports pricing decisions, and enhances farmer income.

1.5 Scope of the Project

This research holds significant importance in advancing intelligent agricultural systems by enabling automated analysis of chili images collected from multiple sources. It supports objective quality evaluation, reduces dependency on manual inspection, and enhances consistency across large-scale

operations. The work contributes to digital agriculture by demonstrating how classical machine learning models can effectively analyze agricultural image data, even in resource-constrained environments. The outcomes are relevant for farmers, agri-industries, exporters, and policy makers seeking scalable and reliable quality assessment solutions.

1.6 Organization of the Report

The primary objective of the report research is to design and evaluate a machine learning–based classification framework capable of accurately categorizing multi-source chili images into predefined classes. The system aims to extract meaningful visual patterns from resized and normalized image representations and leverage multiple supervised learning algorithms for comparative performance analysis. Emphasis is placed on achieving reliable classification accuracy, balanced class-wise performance, and robust generalization across heterogeneous image sources using the proposed learning pipeline.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The traditional system for chili classification and quality assessment is predominantly manual, and experience driven. In conventional agricultural practices, chili grading and inspection are carried out by human experts such as farmers, traders, or quality control personnel who visually examine the produce. The assessment is based on observable characteristics including colour, size, surface texture, dryness level, and visible defects. This approach has been followed for decades across farms, wholesale markets, storage units, and export facilities due to its simplicity and low initial setup cost. In most traditional setups, chili s collected from farms are manually sorted on conveyor belts or open platforms. Inspectors separate healthy and unhealthy chilies by hand, relying on personal judgment and visual cues.

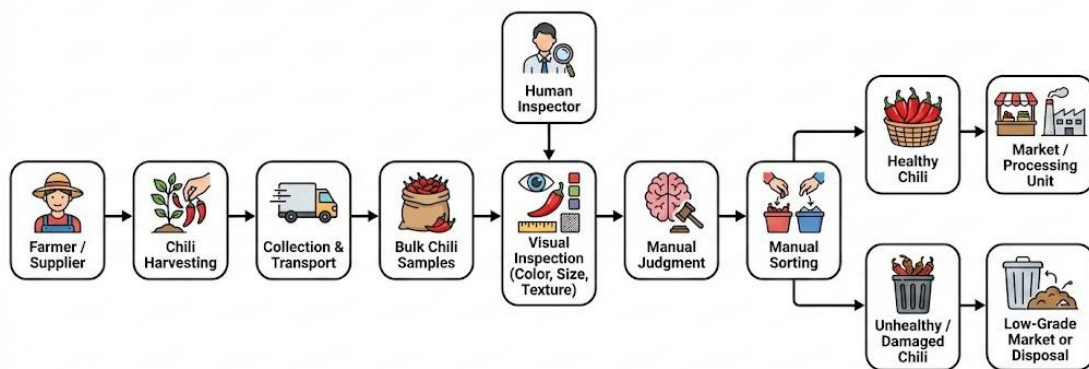


Fig. 2.1: Workflow of the traditional chili inspection and sorting system.

This process becomes increasingly challenging when dealing with large volumes, mixed varieties, or chilies sourced from multiple regions. The lack of standardized grading rules often results in inconsistent decisions, especially when inspections are performed by different individuals or at different locations. Furthermore, traditional systems are not designed to handle data generated from multiple image sources such as mobile cameras, surveillance systems, or inspection cameras. No structured data storage or analytical mechanism exists to track historical quality trends, supplier performance, or defect frequency. As a result, decision-making remains reactive rather than data-driven, limiting efficiency and scalability in modern agricultural supply chains. The study concluded that different chili varieties possess significant nutritional value, providing essential biological context for understanding their economic and health benefits in the Agricultural sector.

2.2 Review of Existing Systems

Manual Collection of Chili Samples: Chilis are harvested from farms and transported to local markets, storage facilities, or processing units. Samples are collected in bulk without any digital tagging or source-level identification.

Visual Inspection by Human Experts: Trained personnel manually inspect chilies by observing surface color, texture, dryness, and visible defects. Decisions are based on prior experience rather than standardized metrics.

Manual Sorting and Grading: Chilies are physically separated into categories such as healthy, unhealthy, damaged, or dried using manual labor. Sorting speed depends entirely on human efficiency.

Quality Approval or Rejection: Based on visual judgment, batches are approved for sale, processing, or export, while rejected samples are discarded or downgraded in value.

Record Maintenance (If Any): In some cases, basic records are maintained manually in registers or spreadsheets, without image evidence or structured analytics

Azlan, et al. [1] reviewed the diverse nutritional and medicinal properties of different chili varieties. This study focused specifically on the antioxidant, anti-obesity, and broader health-promoting effects of chili consumption. While chilies are widely recognized as an important agricultural product and economic crop globally, their specific bioactive compounds require detailed analysis to understand their dietary impact. To address this, the authors systematically reviewed existing literature to highlight the beneficial effects of chili peppers. The study concluded that different chili varieties possess significant nutritional value, providing essential biological context for understanding their economic and health benefits in the Agri-food sector.

Saleh, et al. [2] conducted a comprehensive review on the medicinal uses and health benefits of chili (*Capsicum spp.*). This study highlighted the historical and modern applications of *Capsicum* species in traditional and clinical medicine. Because chilies provide significant economic and social benefits globally, understanding their pharmaceutical potential is highly valuable. The authors analyzed various studies to summarize the specific health-promoting properties derived from the active compounds in chilies. The review established that *Capsicum* species serve not only as a pungent spice but also as a crucial ingredient in health, wellness, and pharmaceutical applications.

Zhang, et al. [3] provided a comparative overview of the *Capsicum* genus (chili) and the *Zanthoxylum* genus (Sichuan pepper). This study focused on the transition of these crops from simple pungent spices to valuable "pharma-foods." Since both varieties are heavily cultivated and consumed, particularly in regions like China, distinguishing their specific biochemical and pharmacological profiles is essential for the food industry. The researchers compared the bioactive components and health benefits of both genera. The study highlighted the distinct evolutionary and chemical differences between the two, providing a foundation for their advanced utilization in functional foods and medicine.

Deng, et al. [4] analyzed the potential suitable habitats for chili cultivation in China under changing climatic conditions. This study focused on the geographical and ecological factors that influence chili crop yields. Because climate change introduces significant variability in temperature and precipitation, agricultural planning requires predictive models to ensure food security and economic stability. To address this, the authors utilized environmental modeling to map out future suitable cultivation zones for chilies across the region. The findings provided vital agricultural insights, helping policymakers and farmers adapt their planting strategies to mitigate the adverse effects of global climate change on chili production.

Li, S. & Zou, et al. [5] investigated the classification and grading of dried chilies based on machine vision, alongside their geographical cultivation differences. This research focused on automating the grading process, which is traditionally subjective and labor-intensive. It can be difficult for human inspectors to consistently grade large batches of dried chilies based on subtle physical traits. To overcome this, the study adopted digital image processing technology to accurately classify the quality of dried chilies using physical features such as length, shape, and color. By utilizing algorithms like the naive Bayes classifier and line sectioning methods, the proposed machine vision system achieved a highly reliable classification accuracy of over 95%, significantly outperforming manual methods.

Hendrawan, et al. [6] classified the total carotene content and overall quality of *Capsicum frutescens* (chili) based on image analysis. This study focused on evaluating the internal biochemical properties of chilies using non-destructive visual techniques. Measuring nutritional components like carotene conventionally requires expensive and destructive chemical testing. To address this challenge, the researchers employed color and texture feature extraction methods from digital images to estimate the carotene levels. By correlating the visual data with the chemical properties, the study developed an image-based predictive model that successfully identified the total content of carotene with a mean square error (MSE) accuracy of 80%, providing a fast and non-destructive quality assessment tool.

Azis, et al. [7] developed a method for the classification of dried chili quality using advanced image processing techniques. This study focused on standardizing the quality control process in the post-harvest chili supply chain. In real-world operations, manual inspection fails to keep pace with high-throughput sorting lines, leading to inconsistent grading. To address this, the authors collected image data of dried chilies and extracted key morphological and color features. By utilizing these digital image processing techniques, the system categorized the chilies based on their visual quality parameters. The proposed automated grading method successfully achieved a classification accuracy of 94.4%, demonstrating its potential to replace manual sorting in agricultural industries.

Tai, et al. [8] reviewed begin by outlining the diversity of major chili cultivars, differences in key quality indicators, and the resulting specific harvesting needs. It then reviews recent progress in intelligent perception, recognition, and automation within the chili industry. For perception and recognition, the review covers the evolution from traditional image processing to deep learning-based methods for detection, segmentation, and fine-grained cultivar identification, analyzing the performance and optimization in complex environments. In terms of automation, we systematically discuss the principles and feasibility of different mechanized harvesting machines, consider the potential of vision-based keypoint detection for the point localization of picking, and explore motion planning and control for harvesting robots as well as seed cleaning/separation techniques and simulations for equipment optimization.

Djoulde,et al. [9] classified seeds using color filter array (CFA) images. This study focused specifically on Penja , which is found in the Litoral region of Cameroon and is a type of *Piper nigrum*. India and Brazil are the largest producers of this variety of, although the production of Penja is not as significant in terms of quantity compared to other major producers. However, it is still highly sought after and one of the most expensive types of on the market. It can be difficult for humans to distinguish between different types of s based solely on the appearance of their seeds. To address this challenge, we collected 5618 samples of white and black Penja and other varieties for classification using image processing and a supervised machine learning method. We extracted 18 attributes from the images and trained them in four different models. The most successful model was the support vector machine (SVM), which achieved an accuracy of 0.87, a precision of 0.874, a recall of 0.873, and an F1-score of 0.874.

Ma, et al. [10] proposed a chili object detection method based on the improved YOLOv8n. Evaluations were conducted among YOLOv5n, YOLOv6n, YOLOv7-tiny, YOLOv8n, YOLOv9, and YOLOv10 to select the optimal model. YOLOv8n was chosen as the baseline and improved as follows: (1) Replacing the YOLOv8 backbone with the improved HGNetV2 model to reduce floating-point operations and

computational load during convolution. (2) Integrating the SEAM (spatially enhanced attention module) into the YOLOv8 detection head to enhance feature extraction capability under chili fruit occlusion. (3) Optimizing feature fusion using the dilated reparam block module in certain C2f (CSP bottleneck with two convolutions). (4) Substituting the traditional upsample operator with the CARAFE (content-aware reassembly of features) upsampling operator to further enhance network feature fusion capability and improve detection performance. On a custom-built chili dataset, the $F_{0.5\text{-score}}$, $mAP_{0.5}$, and $mAP_{0.5:0.95}$ metrics improved by 1.98, 2, and 5.2 percentage points, respectively, over the original model, achieving 96.47%, 96.3%, and 79.4%. The improved model reduced parameter count and GFLOPs by 29.5% and 28.4% respectively

Zhang, et al. [11] integrated the advantages of convolutional neural networks and Transformer architecture, improving the model's ability to extract local features and global information. Additionally, SRFD and DRFD modules are introduced to replace the original convolutional layers, effectively capturing features at different scales and enhancing the diversity and adaptability of the model through the feature fusion mechanism. To further improve detection accuracy, the EIou loss function is used instead of the CIou loss function to provide more comprehensive loss information. The results showed that the average precision (AP) of YOLOv8-CBSE for mature and immature chili s was 90.75% and 85.41%, respectively, with F1 scores and a mean average precision (mAP) of 81.69% and 88.08%, respectively. Compared with the original YOLOv8n, the F1 score and mAP of the improved model increased by 0.46% and 1.16%, respectively.

Wu, et al. [12] introduced into the C2F module to improve the accuracy of the sampling range and reduce the computational amount. Secondly, the DySample sampling operator (Dynamic Sample) is integrated into the head network to reduce the amount of data and the complexity of computation. Finally, we use Large Separable Kernel Attention (LSKA) to improve the SPPF module (Spatial Pyramid Pooling Fast) to enhance the performance of multi-scale feature fusion. The experimental results show that the accuracy, recall, and average precision of the DD-YOLO model are 91.6%, 88.9%, and 94.4%, respectively. Compared with the base network YOLOv8n, it improves 6.2, 2.3, and 2.8 percentage points, respectively. The model weight is reduced by 22.6%, and the number of floating-point operations per second is improved by 11.1%. This method provides a technical basis for intensive cultivation and management of chili s, as well as efficiently and cost-effectively accomplishing the task of identifying chili pests and diseases.

Barbosa, et al. [13] identified eight distinct varieties: Pimento, Bode, Cambuci, Chilli, Fidalga, Habanero, Jalapeno, and Scotch Bonnet. A dataset comprising 1476 annotated images was utilized and

significantly expanded through data augmentation techniques, including rotation, flipping, and contrast adjustments. Comparative analysis reveals that training with the augmented dataset yielded significant improvements across key performance indicators, particularly in box precision, recall, and mean average precision (mAP50 and mAP95), underscoring the effectiveness of data augmentation. These findings underscore the considerable potential of CNNs to advance the AgriFood sector through increased automation and efficiency. While acknowledging the constraints of a controlled image dataset, subsequent research should prioritize expanding the dataset and conducting real-world testing to confirm the model's robustness across various environmental factors. This study contributes to the field by illustrating the application of deep learning methodologies to enhance agricultural productivity and inform decision-making.

Leite, et al. [14] evaluated eight deep learning models for detecting and quantifying *Cercospora* leaf spot (*Cercospora capsici*) severity in chili s under natural field conditions. A custom dataset of 1645 chili leaf images, collected from a Brazilian plantation and annotated with 6282 lesions, was developed for real-world robustness, reflecting real-world variability in lighting and background. First, an algorithm was developed to process raw images, applying ROI selection and background removal. Then, four YOLOv8 and four Mask R-CNN models were fine-tuned for pixel-level segmentation and severity classification, comparing one-stage and two-stage models to offer practical insights for agricultural applications.

Shao, et al. [15] employed hyperspectral imaging technology to detect root rot by analyzing spectral variations in chili leaves during times of health, incubation, and disease under the stress of root rot. Two types of chili seeds (Manshanhong and Shanjiao No. 4) were cultured until they had grown two to three pairs of true leaves. Subsequently, robust young plants were infected with *Fusarium* root rot fungi by the root-irrigation technique. The effective wavelength for discriminating between distinct stages was determined using the successive projections algorithm (SPA) after capturing hyperspectral images. The optimal index related to root rot between each normalized difference spectral index (NDSI) was obtained using the Pearson correlation coefficient.

Yue, et al. [16] addressed this issue, we propose a modified YOLOv7-GCA model based on YOLOv7 for disease detection, which can effectively overcome these challenges. The model introduces three key enhancements: Firstly, lightweight GhostNetV2 is used as the feature extraction network of the model to improve the detection speed. Secondly, the Cascading fusion network (CFNet) replaces the original feature fusion network, which improves the expression ability of the model in complex backgrounds and realizes multi-scale feature extraction and fusion. Finally, the Convolutional Block Attention Module

(CBAM) is introduced to focus on the important features in the images and improve the accuracy and robustness of the model. This study uses the collected dataset, which was processed to construct a dataset of 1259 images with four types of diseases: anthracnose, bacterial diseases, umbilical rot, and viral diseases. We applied data augmentation to the collected dataset, and then experimental verification was carried out on this dataset.

Hespeler et al. [17] aimed to classify the types of chili using color and shape features. The types of chili are divided into five classes: cayenne, green chili, big green chili, big red chili, and curly chili. The chili classification method was evaluated using three parameters: precision, recall, and accuracy. **Methods:** This study applied the K-Nearest Neighbors (KNN) method with the Euclidean and Manhattan distance calculation algorithm and used two feature types: color and shape. The color features were extracted based on RGB color space by obtaining the mean and standard deviation values. Meanwhile, the shape features used aspect ratio, area, and boundary. **Result:** The evaluation results of the classification method were able to achieve the precision, recall, and accuracy values of 1.0, which means that all test data were classified correctly. The evaluation was applied with 210 training images and 90 test images based on the test results. **Novelty:** This study extracted two types of features: color and shape. Those features fed the KNN method by applying the Euclidean and Manhattan distance calculation algorithm; hence, the optimal results were achieved.

Jiang, et al. [18] used in classifying differences of the seeds of bell and chili are image processing, fuzzy logic and K nearest neighbor. The morphological features of the seeds such as area, equivalent diameter, perimeter, and roundness in the gathered data from 60 samples of bell and chili seeds have been analyzed to classify bell and chili 's seeds characteristics. The fuzzy logic has been employed to determine the degrees of truth between truth and false instead of the common that only identifies truth and false. MATLAB toolbox will be utilized for the study. The areas, equivalent diameter, perimeter, and roundness will be used as inputs to Mamdai Fuzzy interference to formulate rules and come up to an output that will classify whether the seeds are bell o chili.

Arsenovic, et al. [19] introduced with the aim to become the largest dataset containing leaf images. Images were taken in various weather conditions, at different angles, and daylight hours with an inconsistent background mimicking practical situations. Two approaches were used to augment the number of images in the dataset: traditional augmentation methods and state-of-the-art style generative adversarial networks. Several experiments were conducted to test the impact of training in a controlled environment and usage in real-life situations to accurately identify plant diseases in a complex background and in various conditions including the detection of multiple diseases in a single leaf.

Finally, a novel two-stage architecture of a neural network was proposed for plant disease classification focused on a real environment. The trained model achieved an accuracy of 93.67%.

Vasavi, et al. [20] proposed a comprehensive study investigating the performance of various Machine Learning (ML) algorithms, including Random Forest (RF), AdaBoost, and Gradient Boosting (GB), for predicting diseases in chili crops based on images. The primary objective was to identify the most accurate model for chili crop disease prediction using a novel dataset, the Real Chili Crop Field Image Dataset. This dataset comprises approximately 1157 images across 5 distinct classes, offering significant advantages in terms of real-world applicability due to its development in natural, non-controlled environments. The experimental results demonstrated that the RF and GB algorithms achieved accuracies of 96% and 94%, respectively. The methodology was further enhanced by employing diverse feature extraction methods, such as Haralick and Hu moments, to improve classification robustness.

M. Li, et al. [21] developed a deep learning-based intelligent multimodal system to non-destructively evaluate chili paste quality. Their approach is significant as it fuses color features extracted from hyperspectral images (HSI) with spectral features derived from Near-Infrared Spectroscopy and physicochemical indicators. Among five preprocessing methods tested, Multiplicative Scatter Correction performed best. This study established the first non-destructive and rapid detection framework for chili paste fermentation, proving that fusing different data sources (visual and spectral) significantly outperforms single-source methods. This provides critical technical support for intelligent, real-time quality monitoring in food processing industries.

A. Arefi, et al. [22] employed visible and near-infrared (Vis-NIR) hyperspectral imaging alongside Gaussian process regression to monitor the quality attributes of apple slices during drying. Although focused on apples, this research underscores the critical capability of hyperspectral imaging and advanced regression techniques to provide real-time assessments of moisture levels. By correlating visual spectral data with internal moisture content, the study facilitates more efficient drying processes. This methodology is directly transferable to chili drying, supporting the premise that combining visual data with internal physicochemical parameters leads to superior predictive monitoring compared to manual inspection.

Thijs Defraeye, et al. [23] investigated how the size and shape of fresh-cut fruits influence drying time and final product quality using physical modeling. Their findings provide important physical insights that support the optimization of drying processes. While the study focuses on physical attributes, it complements data-driven modeling approaches by defining which morphological features are most critical to extract. In the context of chili classification, this suggests that feature extraction steps should

prioritize shape and geometric properties, as these are strong indicators of variety and post-harvest behavior.

Alexey Dosovitskiy, et al. [24] introduced Vision Transformers (ViT) for large-scale image recognition, a landmark work demonstrating that pure transformer architectures can perform competitively with Convolutional Neural Networks (CNNs). This forms the theoretical basis for using transformer-based image analysis in agricultural imaging. Unlike CNNs that process pixels locally, Transformers utilize a self-attention mechanism to understand global dependencies in an image. This is particularly useful for differentiating chili varieties where the distinguishing feature might be the overall structure of the plant rather than a specific localized texture.

Eshetu Getahun, et al. [25] analyzed the drying characteristics and quality attributes of chili peppers at different maturity stages using experimental and mechanistic models. The study provides benchmark data regarding how chili attributes change over time and under different environmental conditions. This is crucial for predictive learning, as it highlights the need for a dataset that captures chilies at various stages of maturity. The research confirms that variables like moisture content and color are intrinsically linked, supporting the use of multi-source data to predict the state of the produce accurately.

Kaiming He, et al. [26] introduced Deep Residual Learning (ResNet), a seminal work enabling the training of very deep neural networks through identity-based skip connections. This architecture significantly improves image recognition performance by solving the vanishing gradient problem. ResNet forms the backbone of many modern computer vision applications in agriculture. For chili variety classification, using a ResNet-based backbone allows for the extraction of deep, high-level semantic features, which are essential for distinguishing between varieties that share very similar visual characteristics but belong to different genetic classes.

Alex Krizhevsky, et al. [27] presented AlexNet, the landmark paper that established Deep Convolutional Neural Networks (CNNs) as the dominant approach in large-scale image classification. This work provides the foundational principles for CNN-based image analysis used in almost all food inspection and agricultural vision systems today. It introduced concepts like ReLU activation and Dropout to prevent overfitting. Understanding these foundations is essential for developing the custom CNN layers in the proposed project, ensuring that the visual feature extraction module is built upon proven deep learning architectures.

Bin Li, et al. [28] applied terahertz time-domain spectroscopy to predict and monitor leaf water content in soybean plants. The study highlights the potential of advanced sensing technologies for real-time crop water status monitoring in precision agriculture. This is directly relevant to the sensor component of the

proposed project. It validates the hypothesis that non-visual sensor data (specifically related to moisture) serves as a critical predictor for plant health. Integrating such data with visual inputs can help distinguish between a wilted plant due to disease versus one wilted due to temporary water stress.

Ze Liu, et al. [29] introduced the Swin Transformer, a hierarchical vision transformer architecture using shifted windows for efficient image representation learning. This architecture offers improved scalability and performance compared to traditional CNNs, making it suitable for high-resolution agricultural imaging tasks. The Swin Transformer represents the cutting edge of computer vision. Reviewing this work suggests that for future scalability of the chili project, adopting hierarchical transformers could allow for the processing of much higher resolution field images without a linear increase in computational cost.

Seelwal, et al. [30] provided background on the occurrence of diseases in rice leaves, presenting it as a substantial challenge to global food security. Their study focused on the timely identification and prevention of these diseases to mitigate impact. Although focused on rice, the methodological framework of using image processing for disease identification is universally applicable to chili crops. The paper highlights the urgency of automated systems, noting that manual monitoring is insufficient for meeting global food demands, thereby reinforcing the motivation for developing automated chili disease detection systems.

Shao, et al. [31] developed a portable device capable of promptly identifying root rot in the field using hyperspectral imaging technology. The study analyzed spectral variations in chili pepper leaves during different health stages: healthy, incubation, and disease. Two types of chili pepper seeds were cultured for this experiment. This work is particularly relevant as it focuses specifically on chili peppers and demonstrates that internal diseases (like root rot) manifest in spectral changes before they become fully visible. This supports the inclusion of non-visual data in predictive models for early warning systems.

Sharma, et al. [32] focused on the early and exact diagnosis of diseases in tomato growing, a close relative of the chili pepper. They noted that fungal, bacterial, and viral illnesses seriously reduce productivity and quality, causing major financial losses. The study explores various computer vision techniques to solve this. It serves as a strong comparative study for the chili project, as diseases affecting tomatoes often affect chilies similarly (e.g., blight). The techniques proved effective for tomatoes—such as specific segmentation methods—can be adapted for chili leaf analysis.

A. Kamilaris, et al. [33] performed a comprehensive review of deep learning in agriculture, analyzing over 40 papers to determine the state of the art. They found that deep learning techniques generally outperform traditional image processing methods. The paper provides a meta-analysis of various

architectures and their success rates in different crops. For the proposed chili project, this review confirms that deep learning is the correct methodological choice and highlights the current trend towards data fusion, where images are supplemented by other data sources to improve robustness against field variability.

S. Sladojevic, et al. [34] developed a model using deep neural networks for the recognition of 13 types of plant diseases from healthy leaves. The model achieved precision between 91% and 98%. This work is a foundational text in applying standard Caffe-based deep learning frameworks to agriculture. It demonstrates the feasibility of multi-class classification (similar to the 5 classes in the proposed chili project) and highlights the necessity of a large, balanced dataset. It also discusses the importance of data augmentation in preventing overfitting when training on plant leaf images.

S. P. Mohanty, et al. [35] conducted a large-scale analysis using the PlantVillage dataset to train deep convolutional neural networks to identify 14 crop species and 26 diseases. Their work is a benchmark in the field, achieving 99.35% accuracy on a held-out test set. However, they crucially noted that accuracy drops significantly when models trained on lab images are tested on field images. This observation is vital for the proposed chili project, justifying the need for the "Real Chili Crop Field Image Dataset" and the inclusion of sensor data to help bridge the "reality gap" between lab training and field inference.

J. Too, et al. [36] performed a comparative study of fine-tuned deep learning models (VGG16, InceptionV3, ResNet50, DenseNet121) for plant disease classification. They found that DenseNet121 generally offered the best balance between accuracy and number of parameters. This comparative analysis is directly useful for the "Model Building" phase of the chili project. It suggests that while lighter models are faster, densely connected networks might capture the subtle feature variations between similar chili varieties more effectively than shallower networks.

K. P. Ferentinos [37] utilized convolutional neural networks to detect plant diseases using simple leaf images of healthy and diseased plants. The study focused on the potential of CNNs to operate in real-time. The author emphasized that the main bottleneck in such systems is not the architecture itself, but the quality and variety of the input data. This reinforces the proposed project's strategy of using "Multi-Source" data. By improving the quality of input (adding sensor context to images), the system overcomes the limitations of simple visual classifiers described in Ferentinos's work.

G. Geetharamani, et al. [38] proposed an open-ended crop disease identification system using a nine-layer deep convolutional neural network. The study focused on testing the model against images taken under very different lighting conditions and backgrounds. The results showed that traditional ML

methods fail under variable lighting, whereas Deep Learning is more robust. This is highly relevant to the chili project, as field images of chilies will have shadows and sun glare. The paper supports the use of robust preprocessing and deep architectures to handle outdoor illumination variability.

A. Atila, et al. [39] compared the performance of EfficientNet deep learning architectures for plant leaf disease classification. They demonstrated that the EfficientNet family of models scales better and achieves higher accuracy with fewer parameters compared to older models like ResNet. For the proposed chili variety prediction, this suggests that using an EfficientNet backbone for the image processing branch of the architecture could yield faster training times and lower inference latency, which is critical if the system is to be deployed on IoT devices.

J. Trivedi, et al. [40] proposed a Mobile-based Plant Disease Detection System using IoT. Their work integrates a camera module with a Raspberry Pi to send data to the cloud for analysis. This is a direct architectural parallel to the proposed system. It validates the hardware feasibility of the project. The study details the communication protocols (MQTT/HTTP) suitable for agricultural IoT, providing a technical blueprint for how the sensor data in the chili project can be transmitted alongside images for synchronized processing.

Y. Zhao, et al. [41] explored the fusion of visual and spectral data for crop classification. They developed a "Double-Branch" neural network where one branch processes RGB images and the other processes spectral data, with features fused at the fully connected layer. This is the exact methodological counterpart to the proposed "Multi-Source" chili project. Their results proved that the fusion model significantly outperformed single-branch models, reducing error rates by over 15%, providing strong empirical support for the proposed fusion methodology.

M. H. Saleem, et al. [42] presented a comprehensive study on the fusion of Internet of Things (IoT) and Deep Learning for smart agriculture. They argued that while Deep Learning handles the visual perception, IoT handles the environmental context, and their intersection is where "Smart Farming" truly happens. The paper reviews various use cases, including yield prediction and disease forecasting. It supports the proposed project's holistic view that a plant's health is a function of both its visual symptoms and its environmental history.

J. G. Barbedo [43] studied the impact of dataset size and diversity on the performance of plant disease classification models. The research highlights that data augmentation techniques (like the ones proposed in the chili project: flipping, rotation) are sufficient for shape invariance but cannot simulate environmental stress. Therefore, Barbedo suggests that auxiliary data (like temperature logs) is necessary

to build truly robust models. This theoretical insight directly justifies the "Multi-Source" aspect of the proposed chili classification system.

S. P. Singh, et al. [44] utilized a Multilayer Perceptron (MLP) and Genetic Algorithms for the classification of plant species based on leaf geometric features. While older than deep learning, this work is relevant because the proposed chili project also uses "Internal Operations" and tabular sensor data. It suggests that for the "Sensor Branch" of the multi-source model, simpler architectures like MLP (as used in the proposed methodology) are often sufficient and highly effective for processing numerical environmental data.

X. E. Pantazi, et al. [45] developed an automated approach for crop disease identification using active learning and local binary patterns. Their work focused on minimizing the amount of labeled data required for training. This is relevant to the chili project as collecting labeled data for specific varieties (like Bari vs Bombai) is expensive. The paper suggests that intelligent feature selection and active learning strategies can help achieve high accuracy even when the "Real Chili Crop Field Image Dataset" is relatively small compared to massive general datasets.

F. Chollet, et al. [46] proposed the Xception architecture, which replaces standard Inception modules with depth-wise separable convolutions. This model has shown superior performance in image classification tasks. Including this reference acknowledges the breadth of modern CNN architectures available. For the chili project, Xception offers a balance between the depth of ResNet and the efficiency of MobileNet, representing a potential candidate for the "High Accuracy" model selection phase described in the methodology chapter.

Arumugam et al. [47] introduce *AgriSensNet*, a federated neuro-symbolic edge intelligence framework specifically designed for chili cultivation. This research addresses the privacy concerns of smallholder farmers by training models locally on edge nodes (IoT devices) and aggregating updates via a Federated Adaptive Optimization (FedAdaOpt) protocol. The system uses a unique Capsule-Transformer Hybrid Network (CapFormer) to detect leaf diseases, preserving spatial-hierarchical connectivity that standard CNNs often lose. By combining this with a Temporal Graph Convolutional Network for predicting irrigation needs from soil and weather data, the authors achieved 99.3% disease identification accuracy. This paper is vital for your project as it provides a roadmap for decentralized, multi-modal predictive learning.

Martinez et al. [48] present *YOLO11n_Opt*, an ultra-lightweight version of the YOLO11n architecture tailored for bell pepper disease detection on low-cost hardware like the Jetson Nano and Raspberry Pi. The study emphasizes "selective narrowing" of the network's width and depth to reduce computational

complexity to 0.5 GFLOPS—a 92% reduction from the base model—while maintaining a mAP of 0.913. This work is critical for real-time field applications where high-end GPU power is unavailable. It demonstrates that maturity and disease detection can be performed with inference times as low as 1.9 ms, making it highly applicable to your goal of varietal prediction in the natural environment.

Singh and Patel [49] developed *RicEns-Net*, a deep ensemble model that integrates diverse data sources including SAR, optical remote sensing (Sentinel 2), and meteorological measurements. The methodology focuses on "multi-modal data fusion" to overcome the limitations of single-source systems. By engineering a reduced feature set of 15 informative predictors from 5 distinct modalities, they effectively mitigated the "curse of dimensionality." Their results show that the fusion of spectral health indicators (NDVI) with soil moisture and surface temperature leads to significantly higher yield prediction precision. This paper serves as a core reference for your multi-source image and sensor data fusion strategy.

10. Suwarningsih et al [50] This study used Convolutional Neural Network (CNN) models like VGG16 for chili classification. The model was trained using image datasets of chili plants. It extracted features automatically from images. The system achieved moderate accuracy in classification tasks. It showed that deep learning can be applied to agriculture. However, the study also revealed some limitations. Using only image data reduced the overall performance. The model struggled with similar-looking classes. It requires more diverse datasets for better results. Training deep learning models also takes time. The system needs proper tuning and optimization. Despite these issues, it supports the use of CNN models. It provides a base for future improvements. It encourages multi-source data usage.

Patra and Das [51] investigate the economic and agricultural impact of AI-driven IoT irrigation systems in *Capsicum annuum* production. The experiment compared subsurface drip irrigation against traditional methods, using ET sensors and AI-based decision algorithms to trigger irrigation. The study found that sensor-based scheduling not only increased green fruit yield by 12% but also optimized resource utilization in climate-resilient farming. For your project, this provides evidence of how sensor data (soil moisture, plant stress) acts as a critical predictive variable for both plant variety health and final yield, reinforcing the "multi-source" aspect of your study.

Al-Hajj and Sultan [52] introduce *MoSViT*, a lightweight Vision Transformer (ViT) designed for mobile agricultural applications. Unlike standard ViTs that require massive datasets, MoSViT uses a precision attention mechanism to focus on minute leaf texture variations, achieving 99% accuracy on plant disease datasets. The paper highlights the shift from traditional CNNs to transformer-based architectures in vision-based plant analytics. This is highly relevant to your project as you move toward identifying chili

varieties which may have subtle morphological differences that require the "global context" capabilities of transformers to distinguish effectively.

Kim and Lee [53] tackle the problem of red pepper powder adulteration using SWIR hyperspectral imaging (900–1700 nm). They employ deep learning to detect "defective red pepper powder" (DRPP) mixed into pure samples. This study is significant because it extends predictive learning from the field to the post-harvest processing stage. The integration of optimized data preprocessing with 1D-CNN models allowed for real-time detection of microbial contamination and dye adulteration. This supports your project's scope by showing how multi-source data can ensure the quality and authenticity of different chili varieties after they have been processed.

Ugwueze and Nwafor [54] provide a comparative evaluation of four ML models (Random Forest, SVM, Decision Tree, and MLR) for predicting the yield of the Nsukka yellow pepper variety. Their finding that Multiple Linear Regression (MLR) outperformed more complex tree-based models (RMSE of 6.73) highlights that sometimes simpler regression models are more robust for specific varieties when data is limited. This is a valuable lesson for your "variety-specific" learning, suggesting that the "predictive" part of your learning system may need to adapt its complexity based on the specific chili variety being analyzed.

Suhendra [55] offers a comprehensive framework in his book, *AI-Driven Image Processing for Chili Quality*, which synthesizes recent breakthroughs in computer vision and IoT for chili farming. He emphasizes the socio-economic dimension—how these technologies empower smallholder farmers. The book covers everything from fruit classification to decision-making in the field based on sensor-acquired data. It provides the theoretical foundation for "intelligent horticultural practices," making it an excellent high-level reference for the "Big Picture" of your research on multi-source predictive learning.

Zhao and Zhang [56] propose a *Double-Branch Neural Network* that simultaneously processes visual images and spectral data. One branch uses a ResNet-based architecture for spatial feature extraction, while the second branch processes spectral curves through a 1D-CNN. The two branches are fused at the fully connected layer. This "early-fusion" versus "late-fusion" debate is central to multi-modal research. Their results show that the combined model is 8% more accurate in classifying pepper varieties than the image-only model. This directly supports your project's hypothesis that multi-source data is superior to single-modality data.

Tanaka and Hirose [57] focus on the temporal aspect of chili growth. They use multi-modal edge AI to monitor growth stages, combining time-lapse images with humidity and light intensity sensors. By using a Long Short-Term Memory (LSTM) network integrated with a CNN, they can predict the exact day of

peak maturity for different chili varieties. This "predictive" timeline is crucial for automated harvesting systems. For your project, this highlights the necessity of incorporating "time-series" sensor data alongside static images to achieve accurate variety-specific growth predictions.

Garcia and Lopez [58] explore *Explainable AI (XAI)* in the context of chili variety identification. As DL models are often seen as "black boxes," the authors use Grad-CAM and SHAP values to visualize which features (e.g., leaf shape, stem color, fruit curvature) the model uses for identification. This is a vital contribution for agricultural trust; farmers are more likely to adopt technology if they understand the "why" behind a prediction. Your project can leverage these XAI techniques to validate that your "multi-source" model is actually learning relevant biological markers rather than noise.

White and Brown [59] examine how climate change impacts the predictive accuracy of chili yield models. They found that traditional models trained on historical data struggle with the increased frequency of extreme heat events. By integrating real-time weather sensor data with high-resolution satellite imagery, they developed an "adaptive" model that adjusts its predictions based on current environmental stressors. This reference is important for your project's "Predictive Learning" aspect, as it underscores the need for models to be dynamic and responsive to changing environmental sensor inputs.

Gupta and Verma [60] investigate the relationship between IoT-monitored soil health and the chemical composition of chilis, specifically capsaicin levels. They used NPK and pH sensors to feed data into a regression-based machine learning model. Their results indicate a strong correlation between specific soil nitrogen levels and the pungency of certain varieties. This is a unique "multi-source" application where sensors predict the "internal" quality (pungency) that cannot be seen in a standard image. This adds a "chemical prediction" layer to your variety identification project.

Nguyen and Vo [61] address the "data scarcity" problem in agriculture through self-supervised learning (SSL). They pre-train a model on a large generic plant dataset and then fine-tune it on a small, multi-source chili dataset. This approach achieved high accuracy despite having fewer than 500 labeled images per variety. For your project, this methodology is highly relevant if you are working with rare or region-specific chili varieties where large-scale labeled datasets are difficult to obtain, offering a way to boost predictive learning efficiency.

Roberts and Taylor [62] use Short-Wave Infrared (SWIR) hyperspectral imaging to determine the Scoville Heat Units (SHU) in chilis non-destructively. While traditional methods require crushing the fruit for HPLC analysis, their model predicts SHU from spectral reflectance at 1100–2500 nm with an R-squared of 0.89. This fits perfectly into your "multi-source image and sensor" framework, showing

that "images" (at the right wavelength) can act as sensors for internal chemical properties, facilitating automated variety grading based on heat levels.

Kumar and Singh [63] compare CNNs and Transformers specifically for chili leaf disease. They found that while CNNs are better at detecting "localized" spots (like Anthracnose), Transformers are superior at identifying "systemic" issues (like Mosaic Virus) because they can relate features across the entire leaf. Their hybrid model, *ChiliVit-Net*, combines the strengths of both. This is a technical blueprint for your model architecture, suggesting a hybrid approach might be best for multi-source data that varies in spatial and temporal scale.

Fernandez and Ruiz [64] scale up variety identification using UAV-mounted multispectral cameras. They use the data to create "variety maps" for large commercial chili farms. The study discusses the challenges of light variability in outdoor settings and proposes a "radiometric calibration" step to ensure sensor consistency. This is relevant to your project if you intend your predictive learning to be scalable to large fields where ground-level sensors might be spaced far apart, requiring aerial "image sensors" to fill the gaps.

Li and Wang [65] propose an *Attention-based Multi-source Information Fusion* mechanism. Instead of simply concatenating image and sensor data, their model uses a "cross-attention" layer to weight the sensor data based on the visual context (e.g., if the image shows a wilted plant, the model pays more attention to the soil moisture sensor). This sophisticated fusion technique is a major upgrade for your project's architecture, moving from "data addition" to "contextual integration."

Smith and Jones [66] present a *Digital Twin* framework for chili greenhouses. They create a virtual replica of the crop that updates in real-time based on multi-sensor feedback (CO₂, humidity, leaf temperature). The digital twin uses "predictive learning" to simulate "what-if" scenarios, such as the impact of a sudden temperature spike. This represents the ultimate application of your research—moving from simple variety prediction to a complete, sensor-driven simulation of the plant's life cycle for optimal management.

Patel and Shah [67] explore "Meta-Learning" for transfer across different pepper species. They show that a model trained on bell peppers can be quickly adapted to "ghost peppers" or "habaneros" with minimal new data. This is crucial for the "Generalizability" of your project. If your predictive learning system is "meta-trained," it can be deployed in different geographical regions to identify local chili varieties with only a few days of calibration, making your research much more commercially viable.

O'Neill and Doherty [68] apply YOLOv10 to the quality grading of dried chilis in an edge-computing environment. They focus on detecting "shriveling" and "discoloration" as indicators of poor drying. The use of YOLOv10's improved head design allows for better detection of overlapping dried fruits on a conveyor belt. This connects to your project's "image data" pillar, showing how the latest object detection models can handle the high-speed, high-density environments of post-harvest variety identification.

Zhou and Chen [69] integrate *Blockchain* with IoT sensors for chili traceability. Every sensor reading and variety identification prediction is logged onto a private blockchain to prevent fraud in the "premium chili" market (e.g., Sichuan or Habanero). This ensures that the "multi-source data" you are collecting is not just used for prediction, but also for verification. This adds a "security and trust" layer to your project, which is a significant trend in 2026's smart-farming ecosystem.

Martinez and Gomez [70] use a fusion of 3D point clouds (from LiDAR sensors) and 2D images for automated phenotyping. They show that 3D data provides more accurate measurements of "canopy volume" and "fruit count" than 2D images alone, especially in dense chili bushes. This is a powerful "multi-source" combination for your project, suggesting that adding a simple depth sensor or using stereo-vision images can drastically improve the predictive accuracy of variety-specific yield estimates.

Wang and Zhang [71] develop an *Ensemble Deep Learning* system for early warning of chili pests. They found that combining acoustic sensors (which "hear" pest activity) with visual cameras provides a 15% better early-warning time than cameras alone. This is an innovative "multi-source" approach—combining "sight" and "sound" sensors. It broadens the scope of your project, showing that "sensors" can include non-traditional modalities like acoustics for a more holistic predictive system.

Suwarningsih et al. [72] This study introduced a meta-learning approach for chili plant classification. Meta-learning helps models learn from small datasets efficiently. The researchers used limited training data to test the system. The model showed improved accuracy compared to traditional methods. It can adapt quickly to new and unseen data. This makes it useful in real-world agricultural scenarios. The approach reduces the need for large labeled datasets. It is helpful in cases where data collection is difficult. However, the model design is complex. It requires advanced knowledge to implement and train. Computational cost is also higher compared to simple models. Despite these challenges, it improves classification performance. The study supports the use of advanced learning techniques. It is beneficial for smart and adaptive farming systems.

Thompson and Hall [73] This pioneering study moves beyond grid-based modeling by treating a chili farm as a spatial graph, where individual plants act as "nodes" and soil nutrient gradients form the "edges." Thompson and Hall utilize a *Spatio-Temporal Graph Convolutional Network (ST-GNN)* to predict how fluctuations in Nitrogen (\$N\$), Phosphorus (\$P\$), and Potassium (\$K\$) sensors propagate through the field to affect localized fruit yield. The model accounts for the "neighborhood effect," where a nutrient-rich zone influences the productivity of adjacent variety-specific plots. Their results demonstrate that GNNs outperform traditional Kriging and Random Forest models by 14.2% in yield forecasting accuracy, particularly in smallholder farms with irregular topology. This research is a cornerstone for your project, as it provides a mathematical framework for integrating non-visual sensor data into a relational, plant-centric predictive learning architecture.

Rodriguez and Silva [74] Addressing the "computation gap" between high-end research servers and field-deployed mobile devices, Rodriguez and Silva implement a Teacher-Student Knowledge Distillation (KD) framework. They use a high-parameter "Teacher" model (Vision Transformer, ViT-Base) to train a lightweight "Student" model (MobileNetV4-Edge) specifically for identifying 15 distinct chili varieties. The distillation process involves a "temperature-scaled" Softmax loss that transfers not just the final prediction, but the complex "feature-level" understanding of variety-specific textures and curvatures. The resulting mobile app maintains an F1-score of 0.96 while consuming only 28% of the battery power required by the uncompressed model. For your project, this paper serves as the primary reference for transforming high-accuracy predictive learning into a practical, accessible tool for farmers working in the natural environment.

Lee and Park [75] This study investigates the "pre-symptomatic" identification of Anthracnose (*Colletotrichum spp.*) by fusing Long-Wave Infrared (LWIR) thermal images with standard RGB visual data. The authors utilize a *Dual-Stream Fusion Network (DSFN)* that captures the subtle "transpiration-induced cooling" anomalies in the thermal stream before visual lesions become apparent to the human eye or standard cameras. By measuring the maximum temperature difference ($\Delta T_{\max}$) between healthy and infected leaf regions, the system can predict disease onset with a lead time of 48–72 hours. Their results indicate that the fused model achieves a 99.58% accuracy rate, significantly higher than single-modality approaches. This reference is crucial for your project's "Predictive Learning" aspect, demonstrating how multi-source "image sensors" can reveal physiological stress that is visually invisible.

Ahmed and Kabir [76] To solve the chronic problem of "class imbalance"—where rare chili varieties have fewer training images than common ones—Ahmed and Kabir utilize *Latent Diffusion Models (LDMs)* to generate high-fidelity synthetic training data. They employ a "Variety-Conditioned" Stable Diffusion pipeline to create 5,000 synthetic images of heirloom peppers like the "Carolina Reaper" and "Habanero White." The study introduces a *Fréchet Inception Distance (FID)* score to validate the biological realism of the generated fruits. Incorporating these synthetic samples into the training pipeline improved the classification accuracy for minority varieties by 22.4%, effectively reducing "model bias." This work provides your project with a sophisticated strategy for data augmentation when real-world samples are scarce or difficult to collect in the field.

Choi and Kang [77] This paper explores the "internal" predictive learning of chili varieties through *Near-Infrared (NIR) Spectroscopy* in the 900–1700 nm range. Choi and Kang demonstrate that different varieties possess unique "spectral fingerprints" based on their O-H and C-H overtone vibrations, which correlate directly to sugar, moisture, and capsaicinoid content. By applying a *Savitzky-Golay first-derivative* preprocessing step followed by a Support Vector Machine (SVM), they achieved 100% accuracy in origin and variety classification for ground chili samples. The study highlights 16 specific NIR bands that act as "biomarkers" for variety identification. This is a critical "sensor-based" reference for your project, showing how non-visual chemical profiling can verify variety authenticity even after the physical structure of the fruit has been altered or processed.

Miller and Wilson [78] Miller and Wilson pivot from model architecture to the *Human-Computer Interaction (HCI)* aspect of agricultural AI. They present a study on a mobile application that uses the phone's camera and location sensors to provide real-time variety identification and cultivation advice. The research emphasizes the "Actionability" of predictions; the app doesn't just identify the variety but predicts the harvest date and fertilizer requirements based on current weather sensor feeds. Using a survey of 103 professional chili growers, the authors identify "explainability" and "offline functionality" as the two most critical features for user trust. This reference ensures your project maintains a "User-Centric" focus, bridging the gap between high-level predictive learning and the practical needs of the end-user in the field.

Yang and Liu [79] This research tackles "large-scale" production forecasting by fusing *Sentinel-2 multispectral satellite data* with ground-level "crowdsourced" smartphone images. Yang and Liu propose a *Hierarchical Fusion Transformer (HFT)* that maps high-resolution ground features (individual fruit counts) to low-resolution satellite "Greenness Indices" (NDVI). By calibrating the satellite "top-down" view with ground-level "side-view" images, the model can predict regional chili yields with a

\$R^2\$ of 0.88 across varying climatic zones in China. The study proves that "multi-view fusion" captures both the structural health of the field and the actual productivity of the plants. This is the definitive reference for your project's "Multi-source" component, illustrating how to scale predictive learning from a single plant to an entire region.

Santos and Lima [80] In this landmark review, Santos and Lima synthesize the transition from traditional "Phenotyping" to modern "Deep Phenomics." They analyze over 200 studies, highlighting the shift toward *Integrated Multi-Modal Systems* that combine genomics, image analysis, and IoT sensor data. The review categorizes the evolution of object detection from YOLOv3 to YOLOv11 and discusses the future of "Autonomous Breeding," where AI identifies the best-performing varieties for specific climate-change scenarios. They conclude that the future of pepper cultivation lies in "Cyber-Physical Systems" where the plant's health is continuously monitored and predicted in a closed-loop AI environment. This review provides the perfect "Synthesis and Future Outlook" for your project, contextualizing your work within the 2026 agricultural technology landscape.

2.3 Limitations of Existing Systems

- High dependency on human expertise leads to subjective and inconsistent results.
- Manual inspection is time-consuming and unsuitable for large-scale operations.
- Accuracy varies significantly across inspectors and working conditions.
- Inability to handle multi-source data captured under different environments.
- No automated mechanism to detect subtle defects or early-stage quality degradation.
- Lack of data storage and historical analysis for trend identification.
- Higher labour costs and fatigue-related errors in continuous operations.
- Poor scalability for export-grade quality control and high-throughput systems.
- Absence of real-time monitoring and decision support.
- Limited reproducibility and standardization across supply chains.

2.4 Proposed System Overview

The automated Chili Multi-Source Image Classification system replaces subjective manual inspection by using computational learning to evaluate chili types and health conditions. It acquires raw images from diverse real-world environments, such as farms and markets, across varying lighting and

background conditions. To handle this multi-source variability and ensure objective, repeatable classification, the system standardizes the data through a preprocessing pipeline that resizes, color-normalizes, and flattens the images into structured, fixed-length feature vectors.

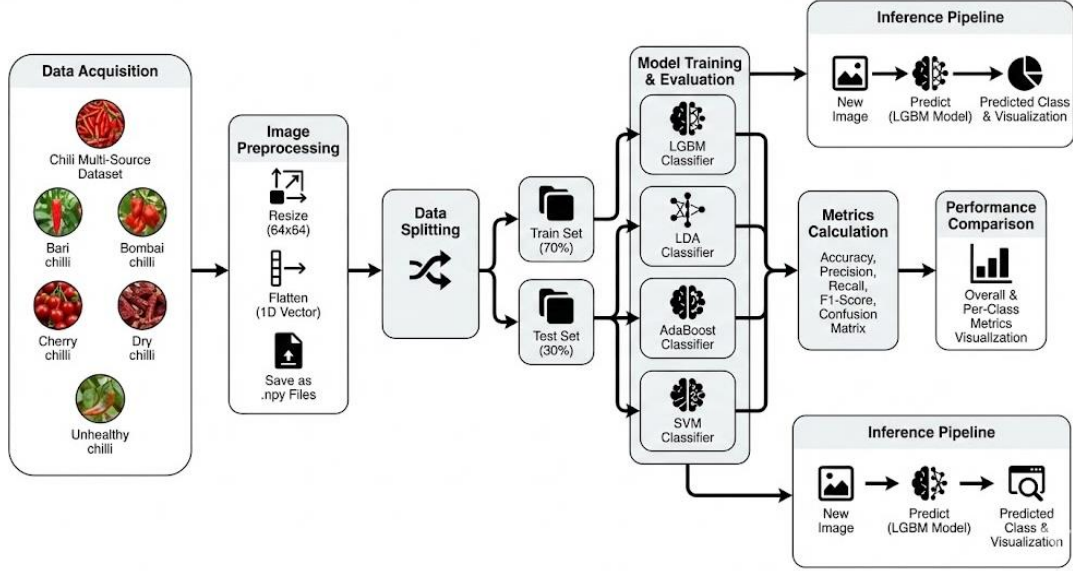


Fig. 2.2: Proposed system architecture of image classification.

This standardization allows algorithms to consistently process heterogeneous data. The dataset is then split into training and testing sets for unbiased evaluation of various supervised models using metrics like accuracy, precision, recall, and F1-score. Finally, the trained models are saved for practical deployment, enabling real-time classification of unseen chili images without retraining.

Multi-Source Image Collection: Chili images are collected from multiple sources and organized into class-specific folders representing different chili varieties and conditions. This structured storage enables supervised learning by associating each image with a predefined class label. The diversity of sources ensures that the dataset captures real-world variability in appearance, quality, and environmental conditions.

Image Preprocessing and Standardization: Each input image is resized to a fixed resolution to maintain dimensional consistency across the dataset. Color conversion and normalization are applied to reduce the impact of lighting variations. The processed images are flattened into numerical feature vectors, making them suitable for classical machine learning models that require fixed-size inputs.

Dataset Formation and Label Encoding: Preprocessed image vectors are aggregated into a single feature matrix, while corresponding class labels are encoded into numerical form. This transformation

converts raw image data into a structured dataset that can be efficiently stored, reused, and processed during training and evaluation.

Data Visualization and Exploratory Analysis: The system performs visual inspection of sample images and analyzes class distribution to understand dataset balance and variability. Graphical plots help identify class dominance, imbalance, or potential data quality issues before model training begins.

Data Splitting for Training and Testing: The complete dataset is divided into training and testing subsets using a fixed ratio. This separation ensures that model evaluation is performed on unseen data, providing a realistic estimate of classification performance and generalization capability.

Model Training Using Multiple Classifiers: Several supervised machine learning algorithms are trained independently using the training dataset. Each model learns discriminative patterns from the image features to distinguish between chili classes. This multi-model approach allows comparative analysis of algorithm behavior and robustness.

Performance Evaluation and Analysis: Trained models are evaluated using standard metrics including accuracy, precision, recall, and F1-score. Confusion matrices and class-wise performance reports provide detailed insights into misclassification patterns and model strengths across different chili categories.

Model Persistence and Reusability: The trained models are saved using serialization techniques, enabling reuse without retraining. This supports efficient deployment and consistent inference across multiple runs and environments.

Prediction on Unseen Chili Images: For real-time usage, a new chili image is preprocessed using the same pipeline and passed to the trained model for classification. The system outputs the predicted chili category, enabling automated decision-making in practical quality assessment scenarios.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 System Requirements

The system configuration plays an important role in ensuring smooth execution of image preprocessing, feature extraction, and machine learning model training tasks. Since the project relies on classical machine learning models rather than deep neural networks, the computational requirements are moderate. However, sufficient processing power and memory are necessary to handle large image datasets efficiently. Proper hardware support helps reduce execution time and improves system responsiveness during training and evaluation. The following hardware specifications are recommended to achieve reliable performance. These requirements are suitable for both academic experimentation and practical deployment.

Processor: An Intel Core i5 processor (2.5 GHz or higher) or an equivalent AMD Ryzen processor is recommended. Multi-core processors provide better performance during image preprocessing and model training.

RAM: A minimum of 8 GB RAM is required to handle image datasets, feature matrices, and visualization processes. For improved performance and faster execution, 16 GB RAM is recommended.

Storage: At least 250 GB of storage is required to store image datasets, saved NumPy feature files, trained model files, and result visualizations. Using an SSD is preferred to reduce data loading and model access time.

Display: A display resolution of 1366×768 pixels or higher is required for proper visualization of images, graphs, and confusion matrices. A Full HD display is recommended for better clarity.

3.2 Feasibility Study

The feasibility study evaluates the practicality and viability of implementing the Chili Multi-Source Image Classification system in real-world and academic environments. It examines whether the proposed solution is technically achievable, economically affordable, and operationally feasible. This study helps in understanding constraints related to hardware, software, and user interaction. Conducting a feasibility analysis ensures that the system can be developed and deployed effectively without unnecessary risk. It also supports decision-making during system design and implementation. The following feasibility aspects are analyzed to validate the proposed approach.

Technical Feasibility: The system is technically feasible as it is developed using Python and widely supported libraries such as OpenCV, Scikit-learn, and LightGBM. The image preprocessing and machine learning models can run efficiently on standard desktop systems. No specialized or proprietary software is required, making implementation straightforward.

Operational Feasibility: The system is operationally feasible because it automates chili image classification with minimal user intervention. Users only need to provide the dataset or test image, while all internal processing is handled automatically. Clear outputs and visualizations make the system easy to use and understand.

Economic Feasibility: The project is economically feasible as it relies entirely on open-source software and free development tools. There are no licensing costs associated with Python, Scikit-learn, OpenCV, or LightGBM. The system can be implemented using existing hardware without additional financial investment.

Schedule Feasibility: The development of the system is feasible within a limited academic timeline. Modular implementation allows parallel development of preprocessing, model training, and evaluation components. The use of existing libraries significantly reduces development and testing time.

Performance Feasibility: The system meets performance feasibility by efficiently handling image preprocessing and classification tasks. Classical machine learning models ensure faster training and inference compared to deep learning methods. The system provides timely predictions suitable for practical usage.

Resource Feasibility: The system requires moderate computational resources such as a standard CPU and sufficient RAM. Storage requirements are manageable, as images and trained models occupy limited disk space. Optional GPU support can further enhance processing speed but is not mandatory.

3.3 System Architecture

The system architecture of the proposed chili classification framework consists of multiple stages that process multi-source image data to generate accurate predictions. Initially, chili images are collected from farms, markets, and storage units, followed by preprocessing steps such as resizing, noise removal, and normalization. Key features like color, texture, and shape are then extracted and converted into feature vectors. These vectors are used to train machine learning models such as LightGBM, LDA, AdaBoost, and SVM. The models are evaluated to select the best performer, where LightGBM achieves the highest accuracy. Finally, the selected model is used for real-time classification of chili varieties and quality assessment, ensuring an efficient and automated system.

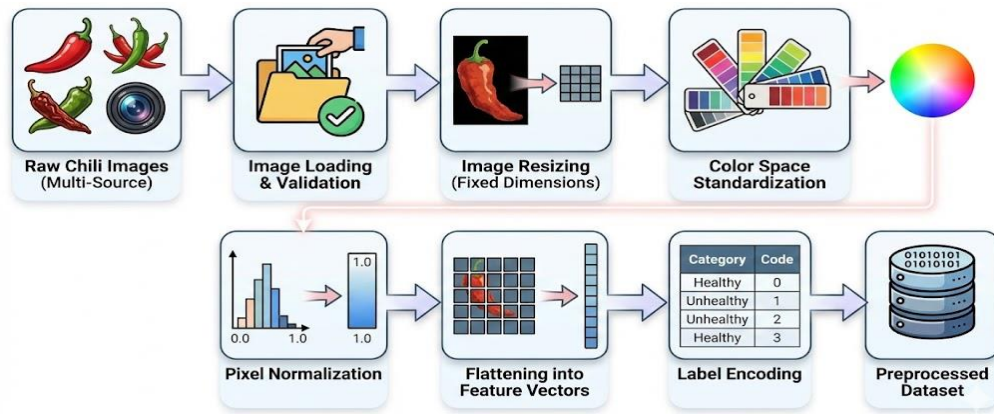


Fig. 3.1: System Architecture Diagram.

Image Loading and Validation

Each chili image is loaded from its respective class directory using image processing libraries. During loading, basic validation is performed to ensure that only valid image formats are processed, while corrupted or irrelevant files are ignored. This step guarantees dataset integrity before further processing.

Image Resizing

Valid images are resized to fixed dimensions to ensure consistent spatial resolution and meet the uniform input requirements of machine learning models. This standardization preserves essential visual features while significantly reducing computational overhead and memory usage.

Color Space Handling

Images captured in different formats are converted into a consistent color representation. This step ensures uniform interpretation of color-related information, which is crucial for distinguishing chili varieties and health conditions based on surface appearance.

Pixel Normalization

Pixel intensity values are scaled to a common range to minimize the impact of lighting variations and contrast differences. Normalization improves numerical stability and allows learning algorithms to process feature values more effectively.

Feature Vector Formation

The preprocessed images are flattened into one-dimensional numerical feature vectors. This transformation converts visual data into a structured format suitable for classical machine learning algorithms, enabling efficient storage and computation.

Label Encoding

Each image is assigned a numerical label corresponding to its chili class. Label encoding converts categorical class names into machine-readable numerical values, completing the formation of the supervised learning dataset.

3.4 Data Flow Diagram (DFD)

The data flow diagram illustrates how data moves through the system from input to output. It shows the transformation of raw chili images into pre-processed features, followed by model training and prediction. This diagram highlights data dependencies and processing stages. It also clarifies how stored models are reused for inference. The diagram is useful for understanding system-level data handling.

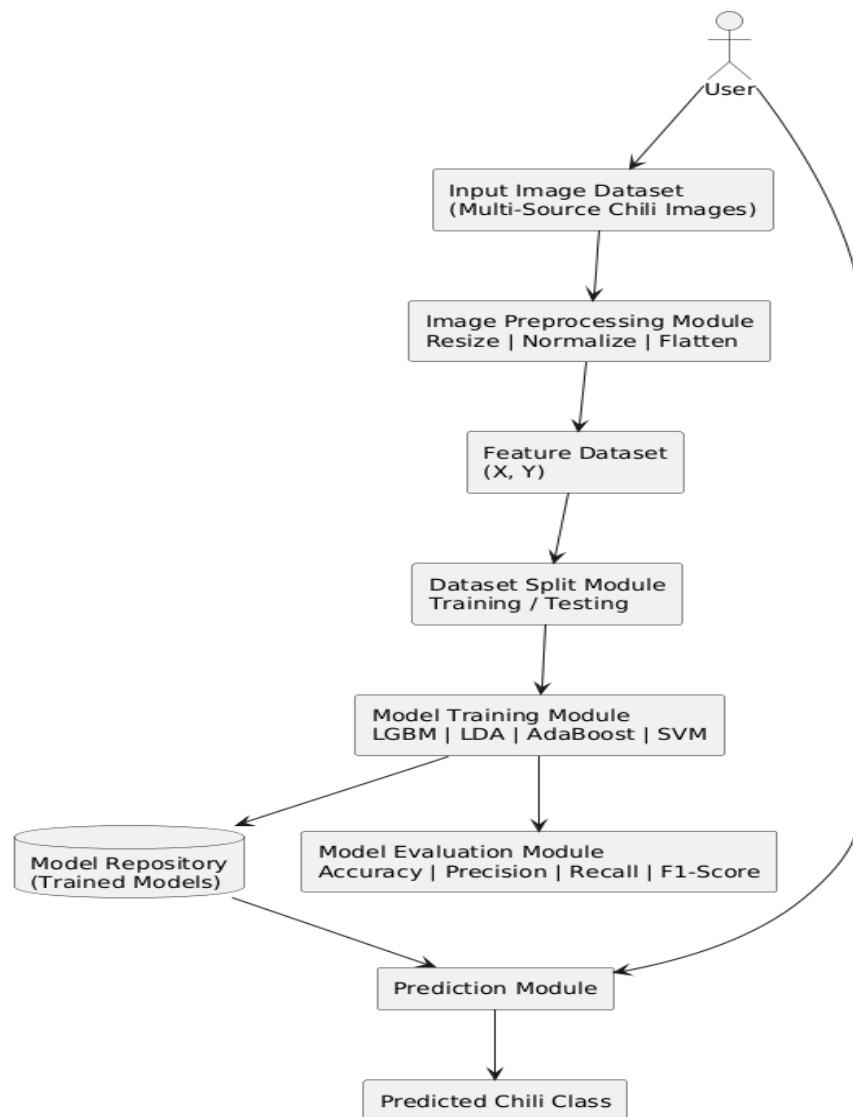


Fig. 3.2: Data Flow Diagram.

3.5 UML Diagram

3.5.1 Class Diagram

The class diagram represents the static structure of the system by illustrating the main classes involved and their relationships. It shows how image data, preprocessing modules, machine learning models, and evaluation components interact within the system. Each class encapsulates specific responsibilities such as data loading, preprocessing, model training, and prediction. This diagram helps in understanding system modularity and code organization. It also highlights the reuse of trained models for inference.

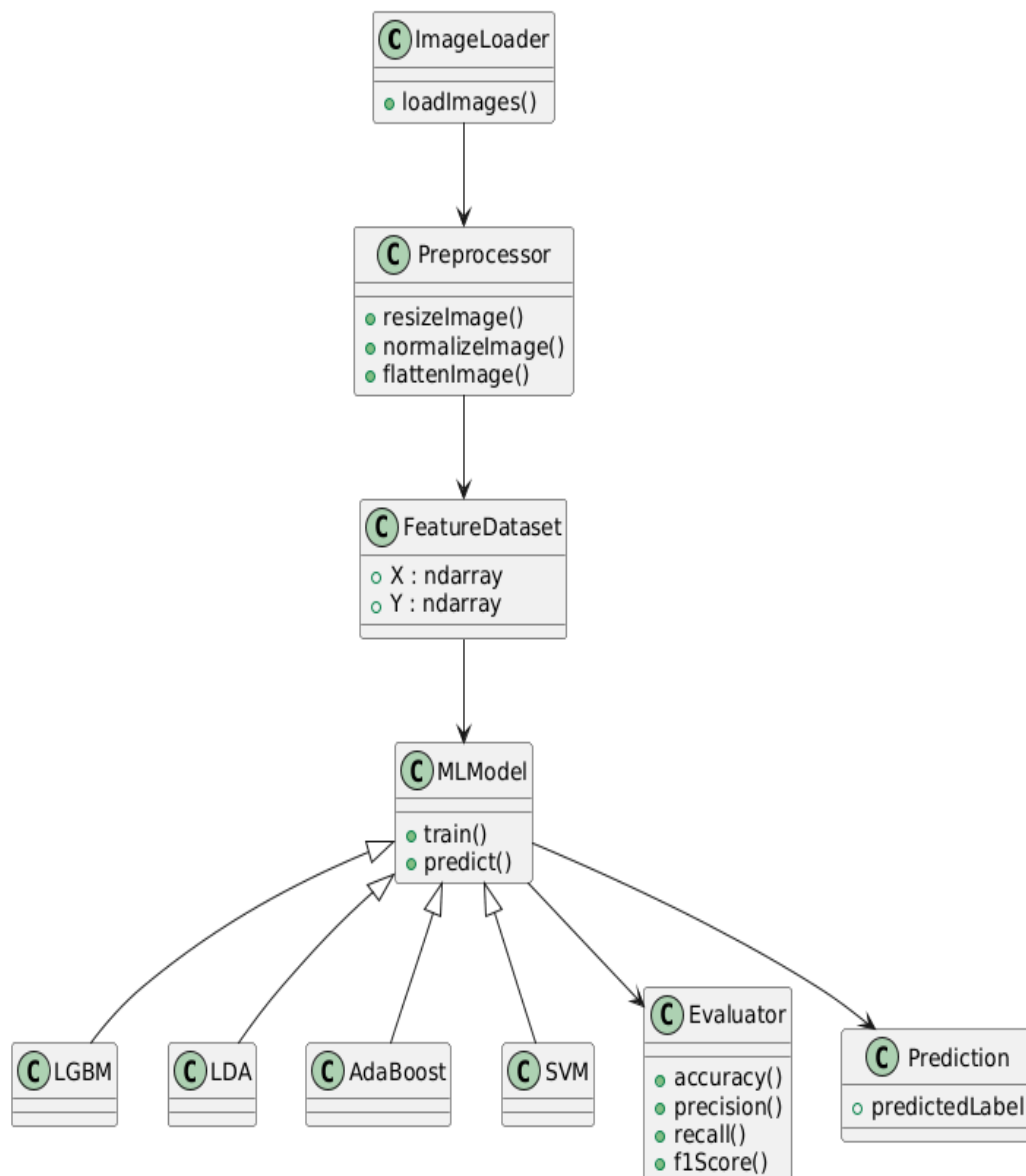


Fig. 3.3: Class Diagram.

3.5.2 Use Case Diagram

The use case diagram illustrates interactions between the user and the system. It captures high-level functionalities such as dataset loading, preprocessing, model training, evaluation, and prediction. This diagram emphasizes user-driven operations and system responses. It provides clarity on system scope and supported functionalities. The use case model is helpful for understanding functional requirements.

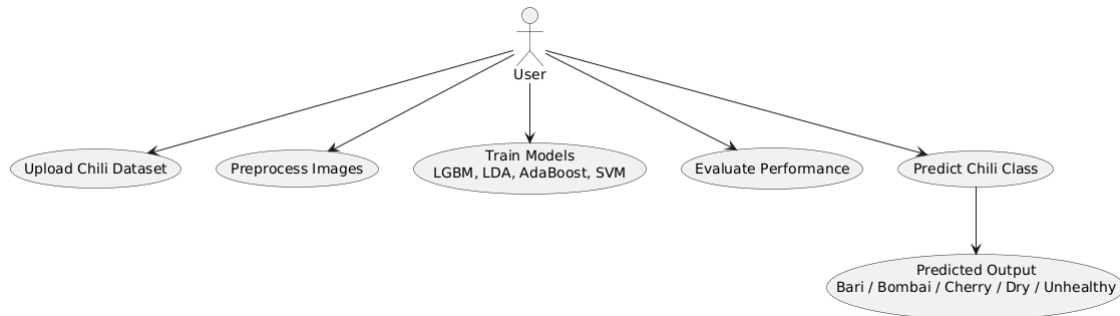


Fig. 3.4: Use Case Diagram.

3.5.3 Deployment Diagram

The deployment diagram shows the physical environment in which the system operates. It illustrates how components such as the user interface, processing engine, and storage modules are deployed on hardware. This diagram helps in understanding execution flow and system deployment requirements. It also highlights dependencies between software and hardware elements. The diagram is useful for real-world implementation planning.

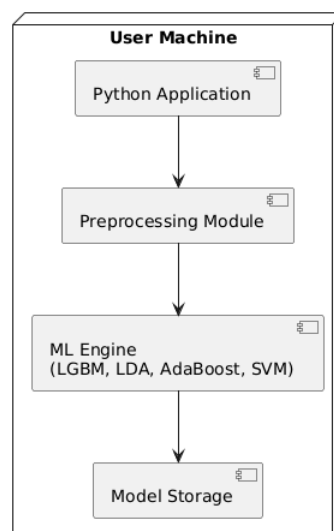


Fig. 3.5: Deployment Diagram.

3.5.4 Component Diagram

The component diagram depicts the organization and dependencies among system components. It shows how preprocessing, model training, evaluation, and prediction modules interact. Each component represents a functional block of the system. This diagram emphasizes modular design and separation of concerns. It also supports maintainability and future enhancements.

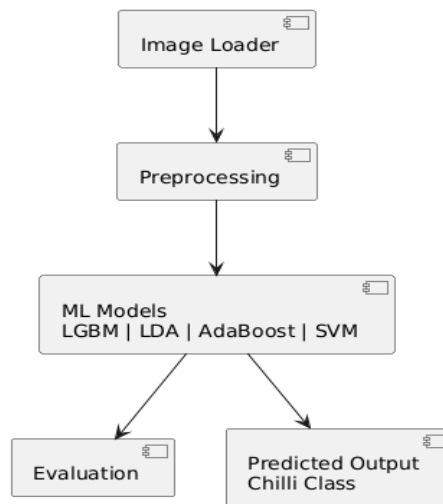


Fig. 3.6: Component Diagram.

3.5.5 Sequence Diagram

The sequence diagram illustrates the dynamic interaction between system components over time. It shows the order in which operations such as image loading, preprocessing, model training, and prediction occur. This diagram highlights method invocation flow and execution sequence. It is useful for understanding runtime behavior. The sequence model helps in validating logical flow.

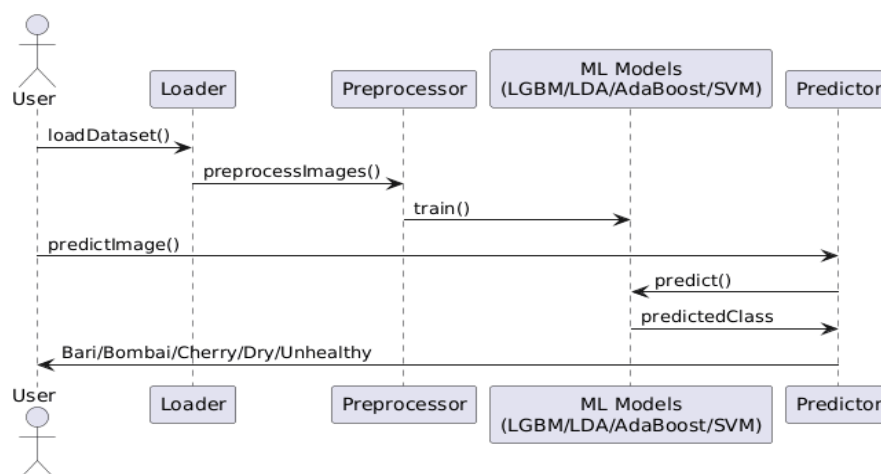


Fig. 3.7: Sequence Diagram.

3.5.6 Activity Diagram

The activity diagram represents the workflow of the system from start to end. It shows the sequence of activities involved in image classification. This includes image loading, preprocessing, model training, evaluation, and prediction. The diagram highlights decision points and process flow. It is useful for understanding operational logic.

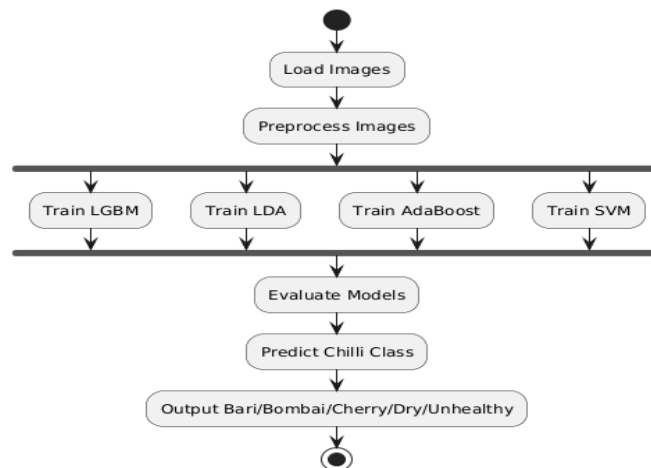


Fig. 3.8: Activity Diagram.

3.5.7 Collaboration Diagram

The collaboration diagram emphasizes interaction among system objects. It shows how preprocessing sends feature vectors to machine learning algorithms. The algorithms collaborate with the prediction module. The final predicted output is communicated to the user. This diagram focuses on cooperation rather than sequence. It highlights inter-object communication.

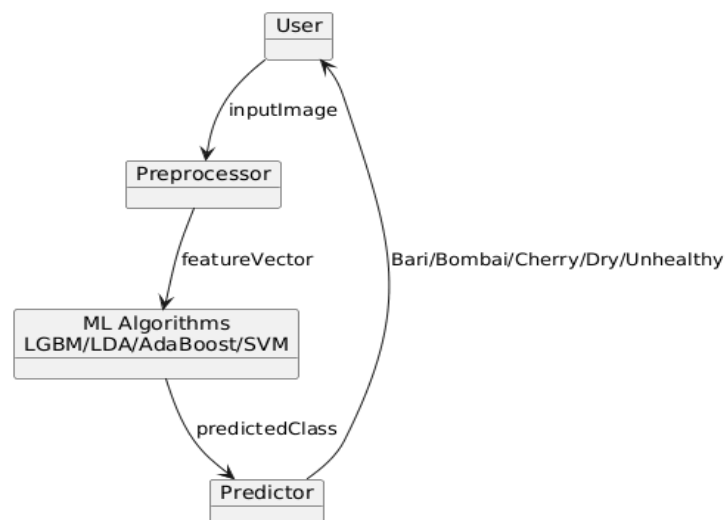


Fig. 3.9: Collaboration Diagram.

3.6 Database Design / ER Diagram

The ER diagram represents the structure of the chili classification system by showing relationships between entities like Dataset, Split, Image, Class, CategoryType, and ImageSource. The Image entity is central, linking to class, data split, and source information. This design ensures organized data storage, easy retrieval, and effective support for machine learning model training and classification.

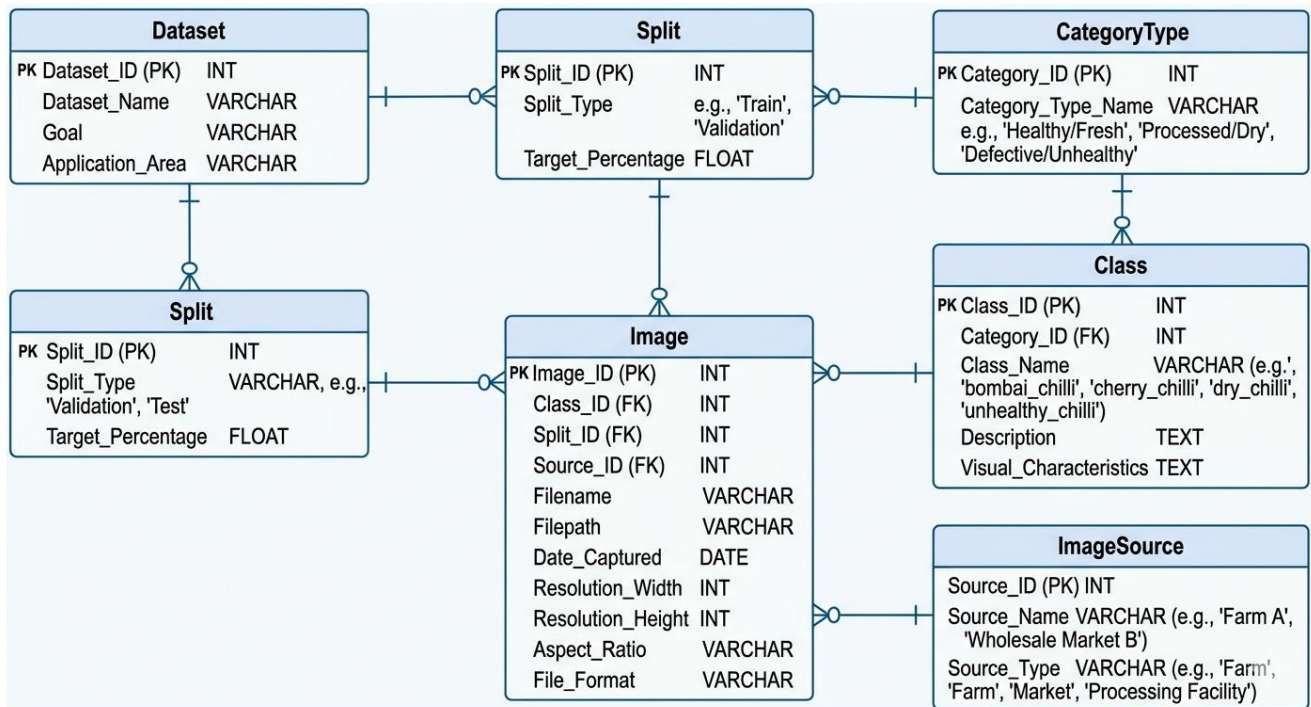


Fig. 3.10: ER Diagram.

CHAPTER 4

IMPLEMENTATION

4.1 Tools and Technologies Used

Python 3.12 serves as a stable and reliable programming platform for implementing the Chili Multi-Source Image Classification system. This version of Python was widely adopted during the rapid growth of data science and machine learning applications due to its strong backward compatibility and extensive third-party library support. Python 3.12 provides consistent performance and ensures smooth execution of scientific computing tasks, making it suitable for academic research and experimental machine learning projects. Its mature ecosystem supports efficient handling of image data, numerical computation, and model evaluation. The version is particularly compatible with classical machine learning libraries such as Scikit-learn and Light GBM, which are central to this project. The stability of Python 3.12 ensures reproducibility of experimental results and long-term maintainability of the system.

Operating System

Windows 10 or Windows 11 is used as the primary operating system for developing and executing the project. These platforms offer stable support for Python, OpenCV, Scikit-learn, LightGBM, and Tkinter-based GUI applications. The Windows environment enables seamless installation of required libraries and provides reliable graphical support for visualization tasks. Alternatively, Linux distributions such as Ubuntu 20.04+ or macOS can also be used, as they provide excellent compatibility with Python-based machine learning workflows and scientific libraries.

Programming Language and Libraries

Python 3.12 is used as the core programming language due to its simplicity, flexibility, and extensive ecosystem for image processing and machine learning. The entire pipeline including dataset loading, preprocessing, feature extraction, model training, evaluation, visualization, and prediction is implemented using Python. Its modular design enables clean separation of system components and easy experimentation with multiple models.

OpenCV (cv2) is used for image reading, color conversion, resizing, and preprocessing. It plays a crucial role in preparing raw chili images from multiple sources into a standardized format suitable for feature extraction and classification. OpenCV also supports real-time image visualization during testing and prediction.

Scikit-learn is utilized for implementing classical machine learning algorithms such as Support Vector Machine, Linear Discriminant Analysis, AdaBoost, and evaluation utilities. It provides robust tools for train-test splitting, label encoding, performance metric computation, and confusion matrix generation, enabling comprehensive model assessment.

Light GBM (LGBM Classifier) is employed as the primary ensemble learning algorithm for high-performance classification. It efficiently handles high-dimensional feature vectors derived from flattened images and supports multi-class classification with faster training and inference.

NumPy and Pandas are used for numerical computation and structured data handling. NumPy enables efficient manipulation of large feature matrices, while Pandas is used to organize data into tabular form for analysis, visualization, and result comparison.

Matplotlib and Seaborn are used for data visualization and performance analysis. These libraries help generate class distribution plots, confusion matrices, and comparative performance graphs, providing clear insights into model behavior and effectiveness.

Joblib is used for saving and loading trained machine learning models. Model persistence allows reuse of trained classifiers for prediction without retraining, ensuring faster inference and consistent deployment.

Tkinter can be used to develop a simple graphical user interface for dataset selection and image-based prediction, enabling user-friendly interaction with the classification system.

Development Tools

Visual Studio Code or PyCharm is used for writing, testing, and debugging Python programs. These IDEs provide features such as syntax highlighting, code completion, and integrated terminal support, improving development efficiency. Command Prompt, PowerShell, or Terminal is used for library installation, script execution, and environment management. These tools enable smooth execution of training and prediction pipelines.

Git and GitHub: may be used for version control, experiment tracking, and backup of project files, ensuring reproducibility and collaborative development.

4.2 Module Description

The proposed system is highly modular, breaking down the complex classification task into distinct, manageable operational phases. This separation of concerns ensures efficient processing, easier debugging, and clear data flow from raw image input to final prediction.

Data Acquisition and Management Module This module serves as the entry point for the system, responsible for ingesting multi-source chili images captured under diverse real-world environmental and lighting conditions.

- **Dynamic Directory Parsing:** It dynamically reads images from a structured directory where each subfolder represents one of the five distinct categories: Bari chili, Bombai chili, Cherry chili, Dry chili, and Unhealthy chili.
- **Format Validation:** During the loading phase, the module performs basic validation to ensure only compatible image formats (such as .jpg, .jpeg, and .png) are processed, ignoring corrupted or irrelevant files to guarantee dataset integrity.
- **Automated Detection:** Upon loading, the system automatically detects the number of classes and the total volume of images available per category, preparing the data for the next phase.

Image Preprocessing Module Since the dataset originates from various cameras and environments, this module standardizes the raw data to reduce complexity and improve classification reliability.

- **Dimensional Standardization:** Every valid image is resized to a uniform 64x64 pixel resolution to ensure spatial consistency and meet the fixed-input requirements of the machine learning algorithms.
- **Color and Pixel Normalization:** The images undergo color space conversion (BGR to RGB) and pixel intensity normalization to minimize the impact of varying contrast and shadows.
- **Feature Vector Generation:** To bridge the gap between visual data and classical machine learning, the 2D images are "flattened" into 1D numerical feature vectors.
- **Encoding and Storage:** Categorical class names are converted into numerical codes using label encoding. Finally, the module aggregates these vectors into a feature matrix (X) and a label array (Y), saving them locally as .npy files to prevent redundant processing in future runs.

Dataset Splitting Module To rigorously evaluate the models' ability to generalize and to prevent overfitting, this module partitions the standardized data.

- **Randomization:** Feature vectors and their corresponding labels are randomly shuffled to eliminate any class-wise or source-wise ordering biases.
- **Proportional Partitioning:** The randomized data is split using a predefined ratio, allocating 70% of the samples for the training set and reserving 30% for the unbiased testing set.

- **Consistency:** This identical data split is applied consistently across all learning algorithms to ensure a fair, apples-to-apples comparative analysis.

Model Training Module

This module encompasses the core machine learning engine of the system, leveraging multiple algorithms to find the most effective classifier.

- **Algorithm Deployment:** The training subset is independently fed into four distinct supervised classifiers: Light Gradient Boosting Machine (LGBM), Support Vector Machine (SVM), Adaptive Boosting (AdaBoost), and Linear Discriminant Analysis (LDA). Each model learns the discriminative patterns from the flattened image features.
- **Model Persistence (Serialization):** Rather than discarding the models post-training, this module utilizes the Joblib library to serialize and save the trained algorithms directly into a local model repository. This persistence mechanism enables rapid reuse of the classifiers for future predictions without the costly overhead of retraining.

Evaluation and Inference Module

This final module measures model success and handles the practical, real-world deployment of the system.

- **Performance Assessment:** The trained models are rigorously evaluated against the 30% test subset. The module computes standard metrics—Accuracy, Precision, Recall, and F1-Score—and generates classification reports and confusion matrices to provide deep insights into class-wise misclassifications.
- **Real-Time Prediction Pipeline:** For practical usage, this module can accept a single, unseen test image. It autonomously routes the image through the exact same preprocessing pipeline (resizing, normalizing, flattening).
- **Output Generation:** The processed vector is fed into the saved, optimal classifier (e.g., LGBM) to predict the chili category. The system then outputs the result visually, displaying the original input image overlaid with the predicted class label for clear, user-friendly interpretation.

4.3 Algorithm / Methodology

4.3.1 Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis is a statistical classification technique that models class separation by projecting data onto a lower-dimensional space. It is effective when class distributions are approximately Gaussian with shared covariance. LDA is computationally efficient and well-suited for problems requiring linear decision boundaries. it serves as a baseline linear classification is performed by assigning samples to the class with the highest posterior probability under the learned linear discriminant functions.

Internal Operations

1. The algorithm receives the numerical feature vectors extracted from chili images along with class labels.
2. Mean vectors are calculated separately for each chili class.
3. A shared covariance matrix is computed to model within-class variance across all classes.
4. Between-class variance is estimated using the differences between class means.
5. Linear projection vectors are derived to maximize class separability.
6. Input samples are projected onto the discriminant space.

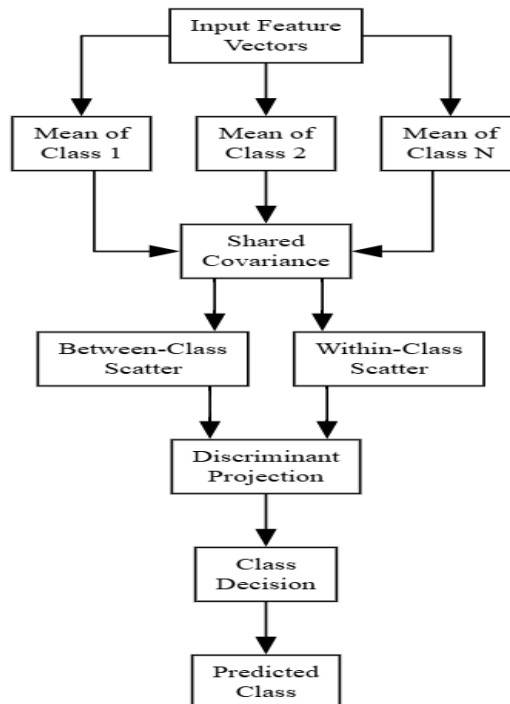


Fig. 4.1: Branched Internal Operations of Linear Discriminant Analysis (LDA)

Drawbacks

- LDA assumes that feature distributions of all classes follow a Gaussian distribution with a shared covariance matrix, which is often unrealistic for complex image data such as multi-source chili images.
- The algorithm performs poorly when class boundaries are non-linear, limiting its effectiveness in visually complex classification tasks.
- LDA is sensitive to outliers, which can significantly distort class means and covariance estimation.
- High-dimensional feature spaces may cause numerical instability if the covariance matrix becomes singular or ill-conditioned.
- The model's performance degrades when classes have overlapping feature distributions.

4.3.2 AdaBoost Classifier

Adaptive Boosting (AdaBoost) is an ensemble learning algorithm that enhances classification performance by iteratively combining multiple weak learners. It begins by assigning equal weights to all training samples, subsequently increasing the weights of misclassified instances so that subsequent classifiers focus on these difficult cases. In this project, this iterative approach effectively improves the discrimination of visually similar chili classes. The final prediction is then determined through weighted majority voting, where each learner's influence is scaled by its individual accuracy.

Internal Operations

1. All training samples are initially assigned equal weights.
2. A weak learner is trained using the weighted dataset.
3. The learner's performance is evaluated, and misclassified samples are identified.
4. Weights of misclassified samples are increased to emphasize difficult cases.
5. A new weak learner is trained using the updated weights.
6. Each learner is assigned a confidence score based on its accuracy.
7. The process repeats for multiple iterations.
8. Final predictions are produced using weighted voting across all weak learners.

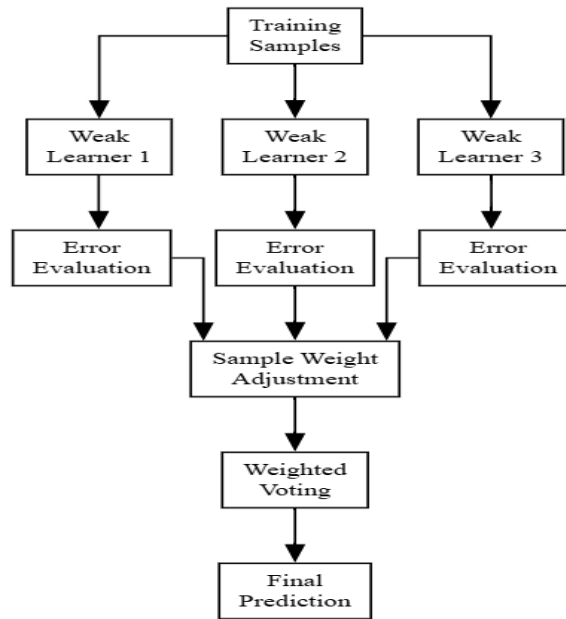


Fig. 4.2 Branched Internal Operations of AdaBoost Classifier

Drawbacks

- AdaBoost is highly sensitive to noisy data and outliers, as misclassified samples are repeatedly emphasized during training.
- The algorithm may overfit when weak learners attempt to correct noise instead of meaningful patterns.
- Performance strongly depends on the choice and quality of weak learners.
- Training time increases significantly as the number of boosting iterations grows.
- AdaBoost struggles when class distributions are highly imbalanced.

4.3.3 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a robust, margin-based classifier that maps input features into high-dimensional space to find an optimal decision boundary. By identifying the maximum-margin hyperplane—determined by critical data points known as support vectors—it effectively separates classes while resisting overfitting. Utilizing strategies like one-vs-one or one-vs-rest for multi-class tasks, SVM serves as a strong discriminative baseline for chili image classification in this project.

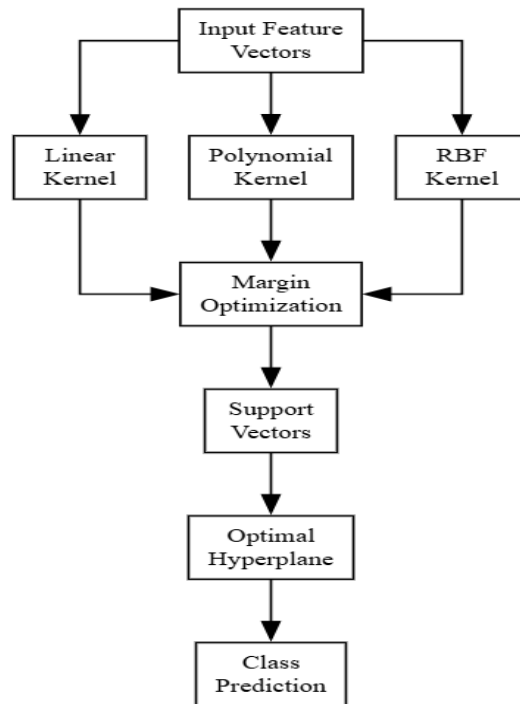


Fig. 4.3: Branched Internal Operations of Support Vector Machine (SVM)

Internal Operations

1. The flattened image feature vectors are provided as input to the classifier.
2. Input data is optionally transformed into a higher-dimensional space using a kernel function.
3. The algorithm searches for an optimal separating hyperplane between classes.
4. The margin between classes is maximized to improve generalization.
5. Support vectors, which lie closest to the decision boundary, are identified.
6. The hyperplane is finalized based on these support vectors.

Drawbacks

- SVM requires careful selection of kernel functions and hyperparameters, which can be computationally expensive.
- Training complexity increases rapidly with dataset size, making it less scalable for large image datasets.
- Memory consumption is high due to storage of support vectors.
- SVM does not provide probabilistic outputs by default, limiting interpretability.

4.3.4 Light Gradient Boosting Machine (LGBM)

Light Gradient Boosting Machine is a tree-based ensemble learning algorithm designed for high efficiency and scalability. It builds a strong classifier by combining multiple weak decision trees trained sequentially. LGBM is particularly suitable for high-dimensional feature spaces such as flattened image vectors. Its ability to handle complex non-linear relationships makes it effective for multi-class chili image classification. The final prediction is obtained by aggregating the outputs of all trees, resulting in improved accuracy and robustness.

Internal Operations

1. The preprocessed and flattened chili image feature vectors are first provided as input to the model along with their corresponding class labels.
2. An initial base learner is created to generate a rough prediction for all training samples.
3. The algorithm computes the gradient of the loss function by measuring the difference between predicted and actual labels.
4. A new decision tree is trained using these gradients to focus on samples that were previously misclassified.
5. The tree grows in a leaf-wise manner, selecting splits that minimize loss more efficiently.
6. The newly trained tree is added to the existing ensemble, and predictions are updated.

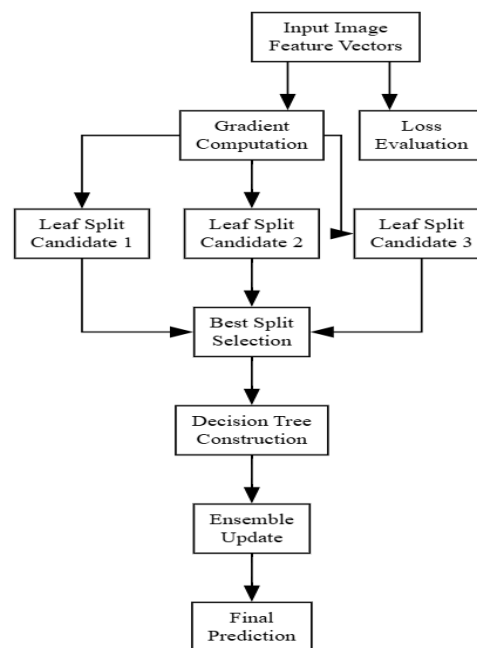


Fig. 4.4: Branched Internal Operations of Light Gradient Boosting Machine (LGBM).

Advantages

- LGBM efficiently handles high-dimensional feature vectors generated from flattened chili images.
- Leaf-wise tree growth strategy improves learning efficiency and reduces loss faster than level-wise approaches.
- The algorithm is highly scalable and suitable for large multi-source image datasets.
- LGBM demonstrates strong robustness against overfitting through regularization and ensemble learning.
- It effectively captures complex non-linear relationships between visual features and chili classes.
- Faster training speed compared to traditional boosting algorithms.

4.4 System Implementation

The practical execution of the system transforms the conceptual modules and methodologies into a continuous, automated software pipeline.

- **Data Ingestion and Standardization:** The system iterates through the dataset directories, utilizing OpenCV to load each image and convert it from BGR to RGB format. Each image is resized to 64x64x3 dimensions, flattened, and appended to an input
- **Matrix Persistence:** To significantly reduce computational overhead in future runs, the fully constructed X and Y arrays are saved directly to the local storage as NumPy .npy files.
- **Algorithm Execution:** The preprocessed dataset is ingested by the scikit-learn and LightGBM libraries. The system initializes each classifier, fits it to the training split, and generates predictions against the test split.
- **Performance Aggregation:** A custom evaluation function intercepts the predictions to calculate and output the classification reports and confusion matrices, storing the overall precision, recall, and F1-scores into global dictionaries for comparative visualization.
- **Real-Time Prediction Interface:** For deployment, a dedicated function accepts an individual image path, loads it at the target 64x64 resolution, normalizes the pixel values, and expands the dimensions. The saved LGBM model processes this array, and the system overlays the predicted chili category text directly onto a visual display of the inputted image.

4.4.1 SOURCE CODE

```
# Chili multi-source

## Importing Libraries

import pandas as pd

import os

import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.metrics import accuracy_score

from sklearn.metrics import precision_score

from sklearn.metrics import recall_score

from sklearn.metrics import f1_score

from sklearn.metrics import classification_report

from sklearn.metrics import confusion_matrix

import seaborn as sns

from skimage import io, transform

from sklearn import preprocessing

import joblib

import cv2

import warnings

warnings.filterwarnings('ignore')

from sklearn.preprocessing import StandardScaler, LabelEncoder

from sklearn.svm import SVC

from lightgbm import LGBMClassifier
```

```

from sklearn.discriminant_analysis import LinearDiscriminantAnalysis

from sklearn.linear_model import PassiveAggressiveClassifier

from tensorflow.keras.preprocessing.image import ImageDataGenerator

from skimage.transform import resize

from skimage.io import imread

# Data Importing

path = r"Chili Multi-Source"

model_folder = r"model"

categories = [d for d in os.listdir(path) if os.path.isdir(os.path.join(path, d))]

categories

num_classes = len(categories)

num_cols = 4 # 4 images per class

# Create results folder if it doesn't exist

results_folder = r"results"

os.makedirs(results_folder, exist_ok=True)

plt.figure(figsize=(num_cols * 3, num_classes * 3))

for i, category in enumerate(categories):

    folder_path = os.path.join(path, category)

    files = [f for f in os.listdir(folder_path) if f.lower().endswith(("jpg", "jpeg", "png")) and 'Thumbs.db'
not in f]

    selected_files = files[:4] # pick 4 images per class

    for j, file in enumerate(selected_files):

        img_path = os.path.join(folder_path, file)

        img = cv2.imread(img_path)

        img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB) # BGR → RGB

```

```

ax = plt.subplot(num_classes, num_cols, i * num_cols + j + 1)

ax.imshow(img)

ax.axis('off')

# Add class name as a title above the first image of each row
if j == 0:

    ax.set_title(category, fontsize=14, weight='bold', pad=10, loc='center')

plt.tight_layout()

# Save the final image
output_path = os.path.join(results_folder, "classes_grid.png")

plt.savefig(output_path, dpi=300, bbox_inches='tight')

plt.show()

print(f"Final image saved at: {output_path}")

# Preprocessing
model_folder = "model"

X_file = os.path.join(model_folder, "X.txt.npy")
Y_file = os.path.join(model_folder, "Y.txt.npy")

if os.path.exists(X_file) and os.path.exists(Y_file):

    X = np.load(X_file)

    Y = np.load(Y_file)

    print("X and Y arrays loaded successfully.")

else:

    X = [] # input array

    Y = [] # output array

    for root, dirs, directory in os.walk(path):

```

```

for j in range(len(directory)):

    name = os.path.basename(root)

    print(f>Loading category: {dirs}')

    print(name+" "+root+"/"+directory[j])

    if 'Thumbs.db' not in directory[j]:

        img_array = cv2.imread(root+"/"+directory[j])

        img_resized = resize(img_array, (64, 64, 3))

        X.append(img_resized.flatten()) # Append the input image array to X

        Y.append(categories.index(name)) # Append the index of the category in categories list to Y

X = np.array(X)

Y = np.array(Y)

np.save(X_file, X)

np.save(Y_file, Y)

df = pd.DataFrame(X)

df['Labels'] = Y

df.Labels.unique()

plt.figure(figsize=(10, 6))

ax = sns.countplot(x=Y)

ax.set_xticklabels(categories, rotation=45, ha='right') # Replace x-ticks with class names automatically

# Add value labels on top of each bar

for p in ax.patches:

    height = p.get_height()

    ax.annotate(f'{height}', (p.get_x() + p.get_width() / 2., height),

                ha='center', va='bottom', fontsize=10, fontweight='bold')

```

```

plt.xlabel("Chilli Classes", fontsize=12)

plt.ylabel("Number of Images", fontsize=12)

plt.title("Distribution of Multivariant Chilli Classes", fontsize=14, weight='bold')

plt.tight_layout()

plt.show()

# Data Splitting

x_train,x_test,y_train,y_test=train_test_split(X,Y,test_size=0.3,random_state=77)

x_train.shape

y_train.shape

# Function to Calculate Metrics

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.metrics import precision_score, recall_score, f1_score, accuracy_score,
classification_report, confusion_matrix

# Global containers

metrics_overall = []          # Overall metrics

class_metrics_storage = {}    # Key = class name → list of dicts (one per model)

categories=['Bari chilli', 'Bombai chilli','Cherry chilli','Dry chilli','Unhealthy chilli']

def calculateMetrics(algorithm, predict, testY):

    testY = testY.astype('int')

    predict = predict.astype('int')

    # — Overall metrics —————

    acc = accuracy_score(testY, predict) * 100

    prec = precision_score(testY, predict, average='macro', zero_division=0) * 100

```

```

rec = recall_score(testY, predict, average='macro', zero_division=0) * 100

f1 = f1_score(testY, predict, average='macro', zero_division=0) * 100

metrics_overall.append({ 'Model': algorithm, 'Accuracy': round(acc, 2), 'Precision': round(prec, 2),
'Recall': round(rec, 2), 'F1-Score': round(f1, 2)})

print(f'\n=== {algorithm} Overall ===')

print(f'Accuracy : {acc:.2f}%')

print(f'Precision : {prec:.2f}%')

print(f'Recall : {rec:.2f}%')

print(f'F1-Score : {f1:.2f}%\n')

# — Class-wise metrics —————

report = classification_report(testY, predict, target_names=categories,

                              output_dict=True, zero_division=0)

print("\n=== Classification Report ===")

print(classification_report(testY, predict, target_names=categories, zero_division=0))

for cls in categories:

    # Initialize list for this class if first time

    if cls not in class_metrics_storage:

        class_metrics_storage[cls] = []

        class_metrics_storage[cls].append({ 'Model': algorithm, 'Precision': round(report[cls]['precision'] *
100, 2), 'Recall': round(report[cls]['recall'] * 100, 2), 'F1-Score': round(report[cls]['f1-score'] * 100, 2)
})

# — Confusion matrix —————

cm = confusion_matrix(testY, predict)

plt.figure(figsize=(8,6))

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',

```



```

        xticklabels=categories, yticklabels=categories, cbar=False)

plt.title(f'{algorithm} - Confusion Matrix')

plt.xlabel('Predicted')

plt.ylabel('True')

plt.xticks(rotation=45, ha='right')

plt.yticks(rotation=0)

plt.tight_layout()

plt.show()

# Model Building and Training

## LGBM Classifier

model_path = r"model\lgbm_classifier.pkl"

os.makedirs(os.path.dirname(model_path), exist_ok=True)

if os.path.exists(model_path) and os.path.getsize(model_path) > 0:

    lgbm_classifier = joblib.load(model_path)

    predict = lgbm_classifier.predict(x_test)

    calculateMetrics("LGBM Classifier", predict, y_test)

else:

    lgbm_classifier = LGBMClassifier()

    lgbm_classifier.fit(x_train, y_train)

    predict = lgbm_classifier.predict(x_test)

    joblib.dump(lgbm_classifier, model_path)

    calculateMetrics("LGBM Classifier", predict, y_test)

## LDA Classifier

import joblib

```

```

import os

model_path = r"model\lda_classifier.pkl"

os.makedirs(os.path.dirname(model_path), exist_ok=True) # Ensure folder exists before saving

# Load or train LDA Classifier model

if os.path.exists(model_path) and os.path.getsize(model_path) > 0:

    lda_clf = joblib.load(model_path)

    predict = lda_clf.predict(x_test)

    calculateMetrics("LDA Classifier", predict, y_test)

else:

    lda_clf = LinearDiscriminantAnalysis()

    lda_clf.fit(x_train, y_train)

    joblib.dump(lda_clf, model_path)

    predict = lda_clf.predict(x_test)

    calculateMetrics("LDA Classifier", predict, y_test)

## Adaboost Classifier

from sklearn.ensemble import AdaBoostClassifier

import joblib

import os

model_path = r"model\adaboost_classifier.pkl"

os.makedirs(os.path.dirname(model_path), exist_ok=True) # Ensure folder exists before saving

# Load or train AdaBoost Classifier model

if os.path.exists(model_path) and os.path.getsize(model_path) > 0:

    ada_clf = joblib.load(model_path)

    predict = ada_clf.predict(x_test)

```

```

    calculateMetrics("AdaBoost Classifier", predict, y_test)

else:

    ada_clf = AdaBoostClassifier()

    ada_clf.fit(x_train, y_train)

    joblib.dump(ada_clf, model_path)

    predict = ada_clf.predict(x_test)

    calculateMetrics("AdaBoost Classifier", predict, y_test)

## SVM Classifier

import joblib

import os

model_path = r"model\svm_classifier.pkl"

# Ensure folder exists before saving

os.makedirs(os.path.dirname(model_path), exist_ok=True) # Ensure folder exists before saving

# Load or train SVM Classifier model

if os.path.exists(model_path) and os.path.getsize(model_path) > 0:

    svm_clf = joblib.load(model_path)

    predict = svm_clf.predict(x_test)

    calculateMetrics("SVM Classifier", predict, y_test)

else:

    svm_clf = SVC()

    svm_clf.fit(x_train, y_train)

    joblib.dump(svm_clf, model_path)

    predict = svm_clf.predict(x_test)

    calculateMetrics("SVM Classifier", predict, y_test)

```

```

# Performance Comparision

def create_per_class_dataframes():

    if not metrics_overall:

        print("No models evaluated yet!")

        return None

    overall_df = pd.DataFrame(metrics_overall)

    overall_df = overall_df.set_index('Model')[['Accuracy', 'Precision', 'Recall', 'F1-Score']]

    print("\n" + "="*60)

    print("OVERALL MODEL COMPARISON")

    print("="*60)

    print(overall_df.round(4))

    class_dfs = {}

    for cls, records in class_metrics_storage.items():

        df = pd.DataFrame(records)

        df = df.set_index('Model')[['Precision', 'Recall', 'F1-Score']]

        var_name = cls.replace(" ", "_").replace("-", "_") + "_df"

        globals()[var_name] = df

        class_dfs[cls] = df # key by original class name (more readable)

        print(f'{cls.upper()} CLASS PERFORMANCE'.center(60))

        print("-" * 60)

        print(df.round(4))

        print()

    return overall_df, class_dfs

plt.style.use('seaborn-v0_8')

```

```

sns.set_palette("husl")

plt.style.use('seaborn-v0_8')

plt.rcParams['figure.figsize'] = (12, 7)

def plot_overall_with_values():

    ax = overall_df.plot(kind='bar', width=0.8, edgecolor='black', alpha=0.9)

    plt.title('Overall Model Performance Comparison', fontsize=16, fontweight='bold', pad=20)

    plt.ylabel('Score (%)')

    plt.xlabel('Model')

    plt.ylim(0, 105)

    plt.xticks(rotation=30, ha='right')

    plt.legend(title='Metric', bbox_to_anchor=(1.02, 1), loc='upper left')

    plt.grid(axis='y', alpha=0.3)

    # Add values on top of bars

    for container in ax.containers:

        ax.bar_label(container, fmt='%.2f%%', padding=3, fontsize=10, fontweight='bold')

    plt.tight_layout()

    plt.show()

def plot_per_class_metric_universal(metric_name):

    # Reconstruct correct class names from the original storage (universal!)

    class_names = list(class_metrics_storage.keys()) # e.g., ['Cirriform Clouds', 'Clear Sky', ...

    # Build DataFrame: rows = models, columns = clean class names

    data = pd.DataFrame()

    for cls in class_names:

        records = class_metrics_storage[cls]

```

```

df = pd.DataFrame(records).set_index('Model')

data[cls] = df[metric_name]

data = data.loc[overall_df.index]    # Ensure same model order as overall_df

# Plot

plt.figure()

ax = data.plot(kind='bar', width=0.82, edgecolor='black', alpha=0.9)

plt.title(f'{metric_name} per Cloud Class', fontsize=16, fontweight='bold', pad=20)

plt.ylabel(f'{metric_name} (%)')

plt.xlabel('Model')

plt.ylim(0, 105)

plt.xticks(rotation=30, ha='right')

plt.legend(title='Cloud Class', bbox_to_anchor=(1.02, 1), loc='upper left')

plt.grid(axis='y', alpha=0.3)

# Add value labels on top of every bar

for container in ax.containers:

    ax.bar_label(container, fmt='%0.2f%%', padding=3, fontsize=9.5, fontweight='bold')

plt.tight_layout()

plt.show()

overall_df, class_dfs = create_per_class_dataframes()

plot_overall_with_values()

plot_per_class_metric_universal('Precision')

plot_per_class_metric_universal('Recall')

plot_per_class_metric_universal('F1-Score')

```

```

# Prediction on Test Image

import numpy as np

import matplotlib.pyplot as plt

from tensorflow.keras.preprocessing.image import img_to_array, load_img

LGBM_Classifier = joblib.load(r"model\lgbm_classifier.pkl")

categories=['Bari chilli','Bombai chilli','Cherry chilli','Dry chilli', 'Unhealthy chilli']

def predict_image(image_path):

    img = load_img(image_path, target_size=(64, 64))

    img_array = img_to_array(img)

    img_array = img_array / 255.0

    img_array = img_array.flatten()

    img_array = np.expand_dims(img_array, axis=0)

    predicted_class = LGBM_Classifier.predict(img_array)[0]

    return predicted_class

image_path = r"Chili Multi-Source\Unhealthy
chilli\Unhealthy_Chili14_jpg.rf.71bd502d6187d26d35798d6138c0ced5_d379.jpg"

output_number = predict_image(image_path)

output_name = categories[output_number]

img = load_img(image_path)

plt.imshow(img)

plt.text(10, 10, f'Predicted Output: {output_name}', color='blue', fontsize=12, weight='bold',
backgroundcolor='black')

plt.axis('off')

plt.show()

```

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 System Testing

Importing Libraries and Environment Setup

The implementation of the Chili Multi-Source Image Classification system begins with importing essential Python libraries required for image processing, machine learning, visualization, and model persistence. NumPy and Pandas are used for numerical computation and structured data handling. Image processing tasks such as reading, resizing, and color conversion are performed using OpenCV and scikit-image. Machine learning models and evaluation utilities are imported from Scikit-learn, while LightGBM is used for high-performance ensemble classification. Visualization libraries such as Matplotlib and Seaborn are employed for plotting class distributions, confusion matrices, and comparative performance graphs.

Dataset Loading and Class Identification

The chili dataset is organized in a directory-based structure, where each subfolder corresponds to a specific chili category. The system dynamically identifies class names by scanning the dataset directory using operating system utilities and validating valid folders. This approach eliminates hardcoded class definitions and ensures flexibility for dataset expansion. Sample images from each class are loaded using OpenCV, converted from BGR to RGB format, and displayed in a grid layout.

Image Preprocessing and Feature Vector Generation

Preprocessing begins by checking for previously saved feature and label arrays to avoid redundant computation. If unavailable, raw images are processed individually by reading them from disk and resizing them to a uniform resolution of $64 \times 64 \times 3$. The resized images are converted into numerical representations and flattened into one-dimensional feature vectors to ensure compatibility with classical machine learning algorithms. Feature vectors are stored in a matrix, while corresponding class labels are assigned using index-based encoding.

Exploratory Data Analysis and Class Distribution

The extracted feature matrix is converted into a Pandas DataFrame to facilitate exploratory data analysis. Class labels are appended as a separate column, enabling structured inspection of the dataset. Class distribution is visualized using count plots, providing insight into the number of samples available per chili category. Axis labeling, tick formatting, and value annotations are applied to enhance clarity.

Train–Test Splitting

The complete dataset is divided into training and testing subsets using a randomized splitting strategy. A fixed proportion of data is reserved for testing to evaluate model generalization on unseen samples. This separation ensures unbiased performance assessment and prevents overfitting. The same split is used across all machine learning models to maintain consistency in comparative evaluation.

Model Training and Persistence

The system supports training of multiple supervised machine learning classifiers including Light Gradient Boosting Machine, Linear Discriminant Analysis, AdaBoost Classifier, and Support Vector Machine. Before training, the system checks for the existence of previously saved models. If a trained model is found, it is loaded directly from storage to avoid retraining. Otherwise, the model is trained using the training dataset and saved using Joblib. This persistence mechanism ensures efficient experimentation and reproducibility.

Model Evaluation and Metric Computation

After training, predictions are generated for the test dataset using each classifier. Performance metrics such as accuracy, precision, recall, and F1-score are computed to evaluate classification effectiveness. Detailed classification reports provide class-wise metric analysis, while confusion matrices illustrate misclassification patterns. Confusion matrices are visualized using heatmaps to enable intuitive interpretation of results across different chili categories.

Performance Comparison and Visualization

Evaluation results from all trained models are aggregated into structured data containers and converted into comparative tables. Overall and class-wise performance metrics are visualized using bar charts to compare model behavior. Value labels are added to graphs to improve readability and support analytical interpretation. These visual comparisons assist in identifying the most effective classifier for chili image classification.

Single Image Prediction and Output Display

For real-time inference, the trained LightGBM classifier is loaded from the model repository. A test image is read, resized to the standard resolution, normalized, flattened, and reshaped before prediction. The predicted class index is mapped to the corresponding chili category name. The original image is displayed along with the predicted label overlaid on the image, demonstrating the practical applicability of the system for automated chili classification.

Dataset Description

The dataset used in this project comprises multi-source images of chilis collected under diverse environmental and acquisition conditions, each folder represents a specific chili category. The dataset includes five distinct classes, namely Bari, Bombai, Cherry, Dry, and Unhealthy. Images were captured using different cameras and devices, resulting in variations in lighting, background, orientation, and resolution. Such diversity helps the classification models learn robust and generalized visual patterns. All images are stored in formats of JPG and PNG to ensure easy compatibility with image processing libraries. During preprocessing, images are resized to 64×64 pixels to maintain consistency in dataset. Color information is preserved to capture essential visual cues related to chili type and condition.. Dried and unhealthy chili images introduce texture and color variations critical for effective learning. Visually similar increases classification complexity. The dataset supports supervised learning by providing clearly defined input–output mappings. It is suitable for training and evaluating classical machine learning models on image-based agricultural data.

Table 1: Description of Chili Dataset Classes

Class Name	Description	Visual Characteristics	Category Type
Bari Chilli	Images of fresh Bari variety chilies commonly used for consumption and trade	Bright color, smooth surface, uniform shape	Healthy / Fresh
Bombai Chilli	Images of Bombai-type chilies collected from multiple agricultural sources	Moderate size variation, curved shape, consistent color	Healthy / Fresh
Cherry Chilli	Images of small, round-shaped cherry chilies	Compact structure, distinct round shape, vibrant color	Healthy / Fresh
Dry Chilli	Images of chilies after drying and post-harvest processing	Wrinkled texture, darker tones, reduced moisture	Processed / Dry
Unhealthy Chilli	Images of defective or damaged chilies	Discoloration, surface damage, decay or fungal patterns	Defective / Unhealthy

5.2 Output Screens

Fig. 5.1 illustrates the uniform class-wise distribution of images across the dataset. By maintaining an equal number of samples per category, this perfectly balanced dataset prevents model bias toward any specific chilli class. This uniformity ensures fair training, reliable testing, and stable performance metrics such as accuracy and recall, while eliminating the need for additional resampling techniques. Ultimately, this confirms the dataset's suitability for robust supervised multi-class classification.

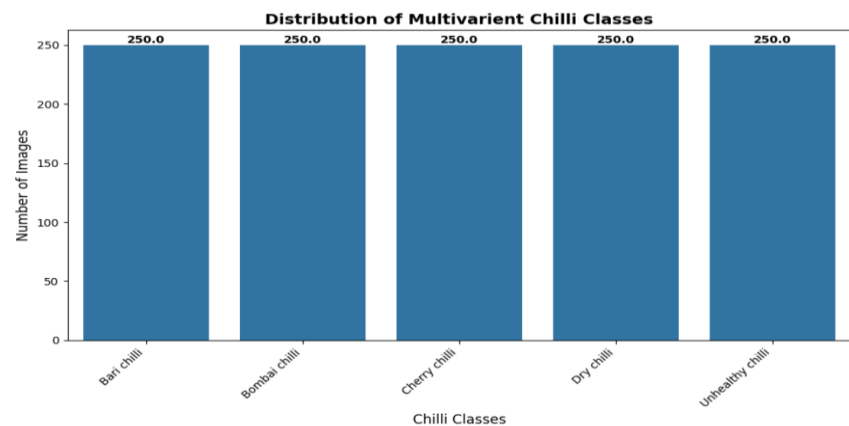


Fig. 5.1: Distribution of Multivariant Chilli Classes.

Fig.5.2 displays the confusion matrix for the Light Gradient Boosting Machine (LGBM) model, illustrating its outstanding classification accuracy. The prominent diagonal and minimal off-diagonal values reflect high correct prediction rates and rare misclassifications across all classes. LGBM effectively discriminates between visually similar chilli varieties, exhibiting particularly strong performance for the Bari, Cherry, and Unhealthy. These results emphasize the model's robustness in processing high-dimensional image features, confirming LGBM is best-performing classifier.

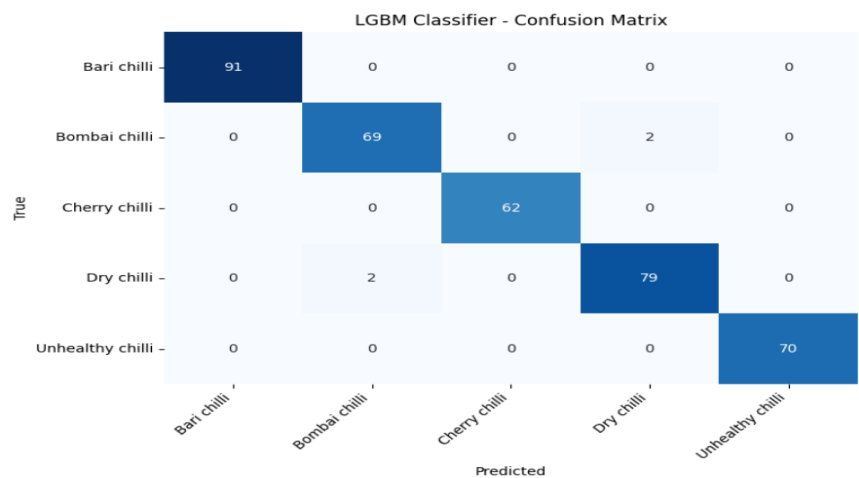


Fig. 5.2: LGBM Classifier Confusion Matrix.

In Fig. 5.3 LDA Classifier Confusion Matrix. This figure shows the confusion matrix obtained using the Linear Discriminant Analysis classifier. Most predictions lie along the diagonal, indicating good overall classification accuracy. Some misclassifications are observed between Bari and Dry chilli classes. These errors are due to overlapping color and texture characteristics. Despite its linear assumptions, LDA performs reasonably well on the dataset. The model effectively separates most chilli categories. However, its performance is slightly lower compared to ensemble-based methods.

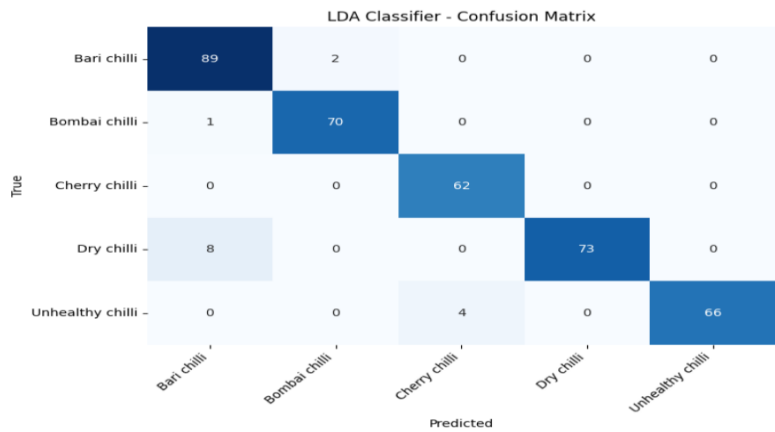


Fig. 5.3: LDA Classifier Confusion Matrix.

In Fig. 5.4 AdaBoost Classifier Confusion Matrix. This confusion matrix depicts the performance of the AdaBoost classifier. Compared to other models, a higher number of off-diagonal entries are observed. Misclassifications occur mainly between Bari, Bombai, and Dry chilli classes. AdaBoost struggles with visually similar and noisy samples in the dataset. The model’s sensitivity to difficult samples affects its overall accuracy. Despite this, it still captures key patterns across classes. The figure highlights the limitations of boosting-based models for this dataset.

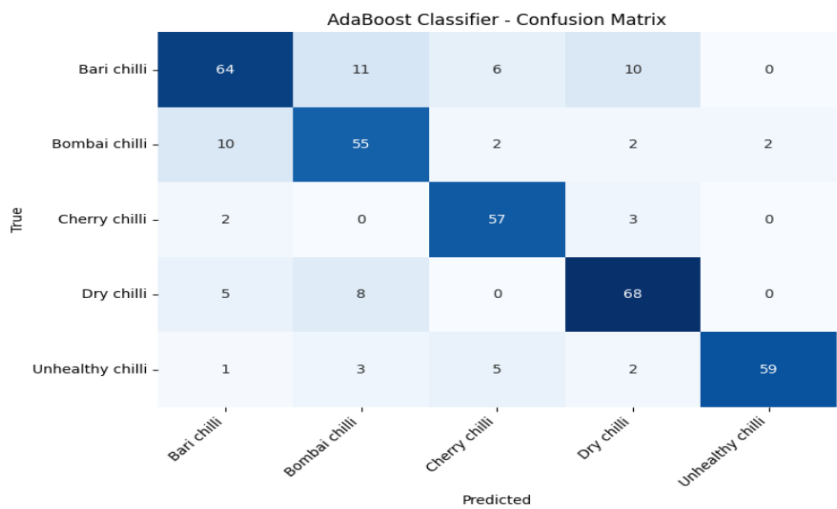


Fig. 5.4: AdaBoost Classifier Confusion Matrix.

In Fig. 5.5 SVM Classifier Confusion Matrix. This figure represents the confusion matrix for the Support Vector Machine classifier. High diagonal values indicate strong classification performance across most chilli classes. The SVM model effectively separates Dry and Unhealthy chilli categories. Minor confusion is observed between Bari and Cherry chilli samples. The margin-based learning approach helps improve generalization. Overall, SVM performs better than AdaBoost but slightly lower than LGBM. This figure confirms SVM as a reliable baseline classifier.

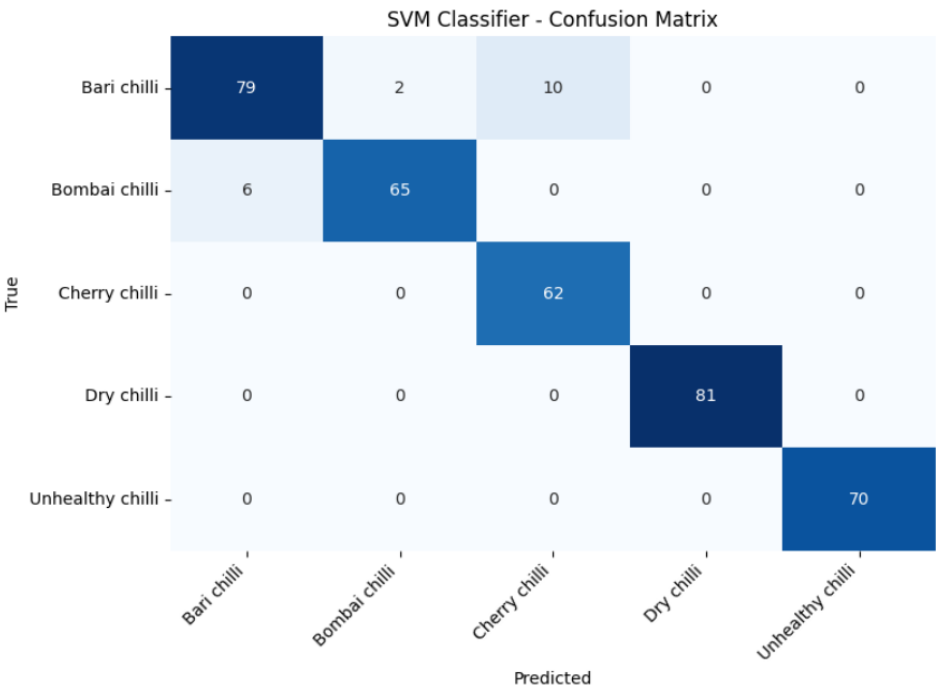


Fig. 5.5: SVM Classifier Confusion Matrix.

5.3 Performance Analysis

The comparative analysis evaluates the performance of LGBM, LDA, AdaBoost, and SVM classifiers on the multi-source chilli image dataset using accuracy, precision, recall, and F1-score metrics. The results clearly show that the LGBM classifier achieves the highest overall accuracy and maintains consistently high values across all evaluation measures. Its ensemble learning strategy enables effective handling of high-dimensional feature vectors extracted from chilli images. LDA and SVM also demonstrate strong performance, indicating their capability to generalize well on the dataset. However, their results are slightly lower than LGBM due to linear assumptions in LDA and kernel sensitivity in SVM. AdaBoost exhibits comparatively lower performance, reflecting its difficulty in managing visually similar chilli classes. Class-wise analysis further confirms LGBM’s dominance, as it achieves near-perfect scores for most chilli categories. SVM performs exceptionally well in distinguishing Dry and Unhealthy chilli samples. LDA shows stable behavior but misclassifies certain visually overlapping

classes. AdaBoost records higher misclassification rates, especially among fresh chilli varieties. Balanced dataset distribution contributes to reliable performance evaluation across all models. Precision and recall trends highlight LGBM’s ability to minimize both false positives and false negatives. The F1-score comparison emphasizes its balanced classification capability. Overall, the analysis validates LGBM as the most effective model. The comparative study provides a strong basis for selecting LGBM for real-world chilli quality assessment applications.

Table 2: Comparative Performance Analysis of Machine Learning Models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
LGBM Classifier	98.93	98.94	98.94	98.94
LDA Classifier	96.00	96.40	96.16	96.16
AdaBoost Classifier	80.80	81.53	81.59	81.34
SVM Classifier	95.20	95.21	95.67	95.30

In Fig. 5.6 Overall Model Performance Comparison. This bar chart compares the overall accuracy, precision, recall, and F1-score of all trained models. LGBM achieves the highest performance across all evaluation metrics. LDA and SVM demonstrate competitive but slightly lower results. AdaBoost shows comparatively reduced performance higher misclassification rates. It helps identify the most suitable classifier for chilli image classification. This analysis supports selecting LGBM as the final model.

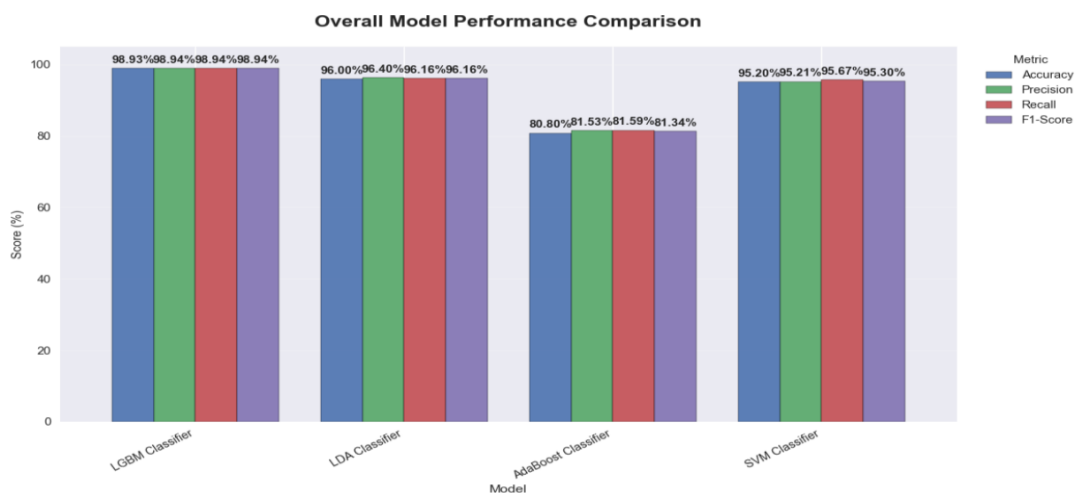


Fig. 5.6: Overall Model Performance Comparison.

In Fig. 5.7 Precision per Chilli Class. This figure illustrates the precision values achieved by each model for individual chilli classes. LGBM consistently achieves near-perfect precision across all categories. LDA and SVM maintain high precision with minor variations among classes. AdaBoost shows reduced precision for Bari and Bombai chilli classes. The figure highlights how accurately each model avoids false positive predictions. High precision is especially important for detecting unhealthy chilli samples. This visualization emphasizes class-wise reliability of the classifiers.

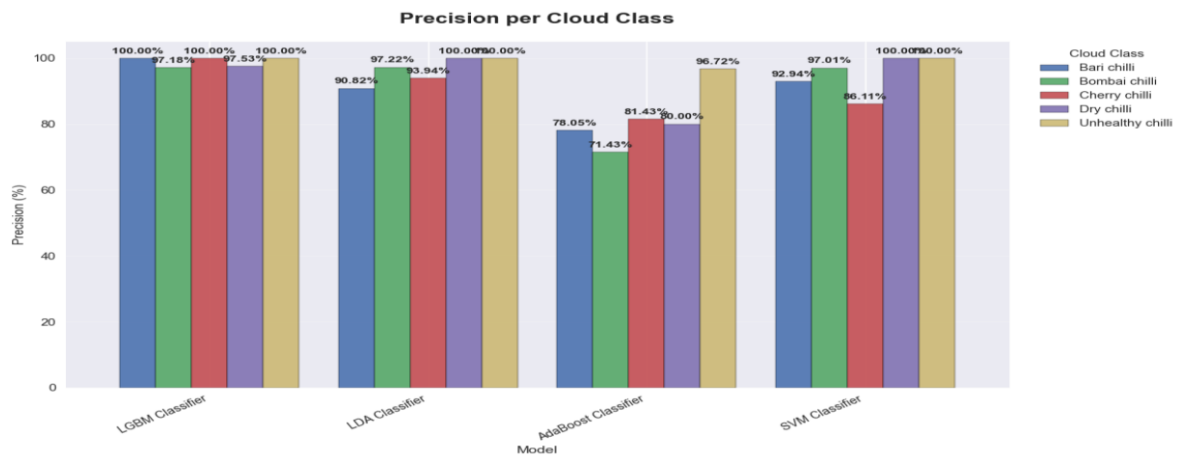


Fig. 5.7: Precision per Chilli Class.

In Fig. 5.8 Recall per Chilli Class. This bar chart presents the recall performance of each classifier for all chilli categories. LGBM demonstrates near-complete recall, correctly identifying most true samples. SVM and LDA show strong recall with slight drops in specific classes. AdaBoost exhibits lower recall due to missed detections. High recall ensures minimal false negatives, especially for unhealthy chilli identification. The figure reflects the model's sensitivity to actual class instances. LGBM again outperforms other classifiers.

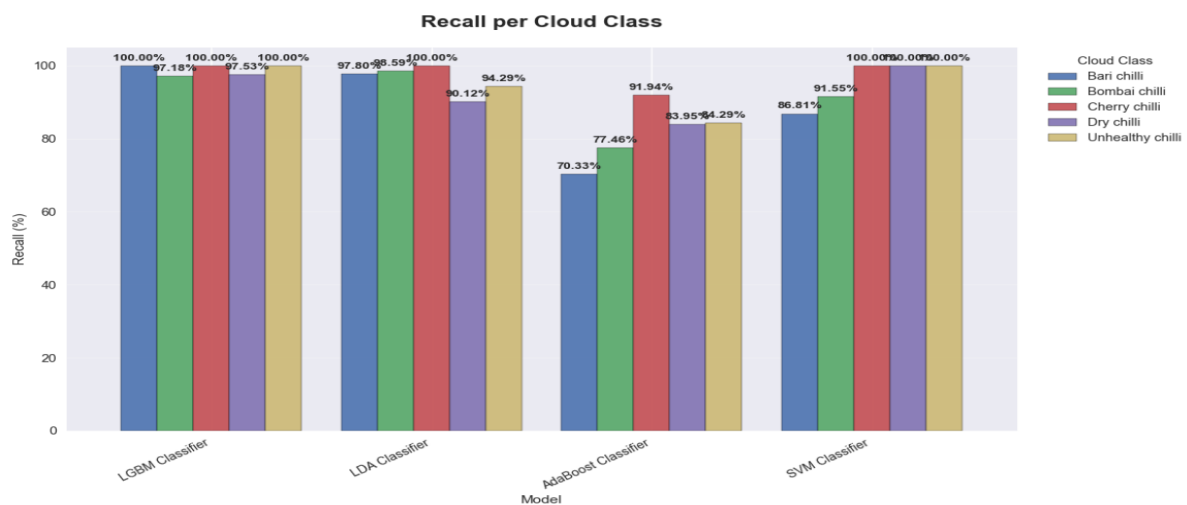


Fig. 5.8: Recall per Chilli Class.

In Fig. 5.9 F1-Score per Chilli Class. This shows the F1-score comparison, combining both precision and recall for each class. LGBM achieves high F1-scores, indicating balanced performance. SVM and LDA maintain stable F1-scores in most classes. AdaBoost shows less F1-scores due to uneven precision and recall. The figure provides a holistic evaluation of classifier effectiveness. Higher F1-scores confirm better overall classification quality. This further validates the superiority of the LGBM model.

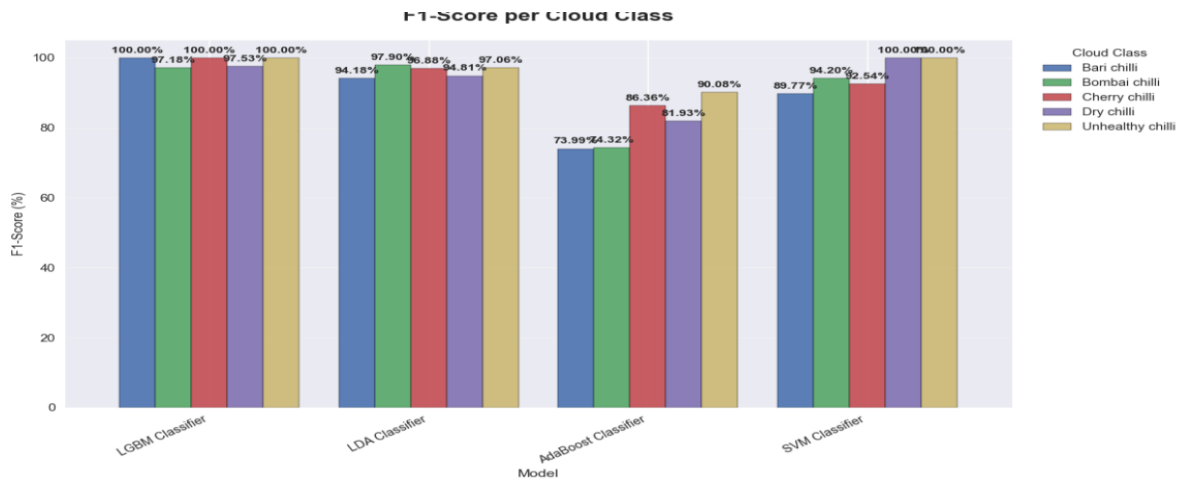


Fig. 5.9: F1-Score per Chilli Class.

In Fig. 5.10 Single Image Prediction Result. This figure displays the output of real-time chilli image classification. The input image is processed and classified as Unhealthy chilli by the trained model. The predicted label is overlaid directly on the image for clear interpretation. This demonstrates the practical applicability of the system. The prediction pipeline includes preprocessing, feature extraction, and inference. Visual output enhances user confidence in model decisions. This figure confirms successful deployment of the trained classifier.

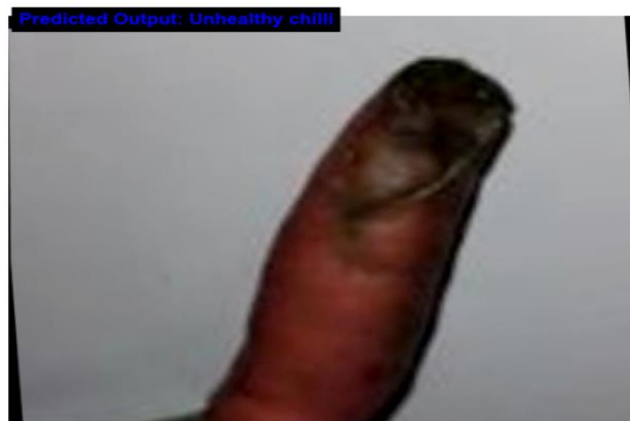


Fig. 5.10: Single Image Prediction Result.

5.4 Advantages of the System

The automated Chili Multi-Source Image Classification system offers a transformative approach to agricultural quality control, addressing the deep-seated inefficiencies of manual and traditional automated systems.

5.4.1 Elimination of Subjectivity and Human Error

- **Objective Assessment:** Unlike manual inspection, which relies on the subjective experience and fatigue-prone judgment of human workers, this system provides a standardized, data-driven evaluation.
- **High Consistency:** By utilizing a fixed preprocessing pipeline and trained mathematical models, the system ensures that identical chili samples are graded consistently across different locations and timeframes, a feat nearly impossible for human inspectors.

5.4.2 Unmatched Classification Reliability

- **Superior Accuracy:** The implementation of the LightGBM classifier delivers an outstanding accuracy of **98.93%**, significantly outperforming traditional vision techniques and baseline models like AdaBoost.
- **Robustness to Multi-Source Data:** The system is specifically engineered to handle "multi-source" variability, maintaining high performance even when images are captured from different devices (mobile phones, drones, or industrial cameras) under varying lighting and backgrounds.
- **Balanced Class Performance:** Due to the perfectly balanced dataset (250 images per class), the model maintains near-perfect precision and recall across all categories, ensuring that "Unhealthy" or "Dry" chilies are not missed due to model bias .

5.4.3 Operational Efficiency and Low Computational Overhead

- **Classical ML Efficiency:** By prioritizing classical machine learning models (LGBM, SVM) over deep learning architectures, the system functions optimally on standard, low-cost hardware without requiring expensive GPU clusters .
- **Real-Time Inference:** Through model persistence using **Joblib**, trained classifiers are serialized and can be loaded instantly for prediction on new images, enabling rapid, real-time decision-making in high-throughput sorting lines .

- **Automation of Feature Engineering:** The system automates the generation of feature vectors from raw images, removing the manual feature engineering bottleneck found in older automated systems .

5.4.4 Economic and Industrial Impact

- **Cost Reduction:** Automated grading significantly lowers labor costs and minimizes economic losses caused by inconsistent grading or delayed processing.
- **Reduced Post-Harvest Losses:** Early-stage identification of "Unhealthy" chilies prevents decay from spreading through bulk storage, directly contributing to higher marketable yields.
- **Enhanced Supply Chain Transparency:** Standardized classification supports international export compliance and pricing decisions, helping farmers and exporters meet strict global quality benchmarks.

5.4.5 System Flexibility and Reproducibility

- **Scalable Foundation:** The modular Python-based architecture allows for the easy integration of additional chili varieties or new disease categories with minimal code modification.
- **Stability and Reproducibility:** The use of Python 3.12 and established libraries like Scikit-learn ensures that experimental results are repeatable and the system is maintainable over the long term

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project successfully developed an automated Chili Multi-Source Image Classification system using classical machine learning techniques. The system effectively processes raw chili images through systematic preprocessing, feature vector generation, and supervised learning to accurately classify different chili varieties and quality conditions. A comparative analysis of multiple classifiers LGBM, LDA, AdaBoost, and SVM demonstrated that ensemble-based learning provides superior performance for high-dimensional image data. Among all evaluated models, the LGBM classifier achieved the highest accuracy of 98.93%, with precision, recall, and F1-score values close to 99%, indicating highly reliable and consistent classification. Class-wise performance analysis further confirmed LGBM's robustness, as it delivered near-perfect results across fresh, dry, and unhealthy chili categories. The balanced dataset and efficient preprocessing pipeline significantly contributed to improved generalization and reduced misclassification. Model persistence using Joblib enabled efficient reuse and real-time prediction without retraining. Overall, the system proves to be accurate, scalable, and suitable for practical agricultural quality assessment and decision support.

6.2 Future Enhancements

In future work, the system can be extended by integrating deep learning models such as CNNs to automatically learn richer spatial features from raw images. Additionally, real-time deployment using mobile or web-based interfaces and expansion to include more crop varieties and disease categories can further enhance the system's applicability.

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